



Washington University in St. Louis

JAMES MCKELVEY SCHOOL OF ENGINEERING



2021

DESIGN REPORT

Acronyms and Nomenclature	4
1. Executive Summary	5
1.1 Overview	5
1.2 Summary of Design	5
1.3 Aircraft Capabilities	5
2. Management Summary	6
2.1 Organization	6
2.2 Schedule	7
3. Conceptual Design	7
3.1 Design Philosophy	7
3.2 Mission Requirements	8
3.3 Design Requirements	9
3.4 Initial Aircraft Designs	10
3.5 Final Design Selection	12
4. Preliminary Design	16
4.1 Aerodynamic Performance	16
4.2 Model Capabilities and Uncertainties	19
4.3 Aerodynamic Analysis	20
4.4 Propulsion Sizing	25
4.5 Initial Performance Estimates	26
5. Detail Design	27
5.1 Dimensional Parameters	27
5.2 Structural Characteristics	28
5.3 System and Subsystem Design	29
5.4 Mass and Balance	33
5.5 Flight and Mission Performance	34
5.6 Drawing Package	35
6. Manufacturing Plan	39
6.1 Manufacturing Methods Considered on Major Components	39
6.2 Manufacturing Process	40
6.2.1 Wing	40
6.2.2 Control Surfaces and Attachments	41
6.2.3 Fuselage	42
6.2.4 Tail	43
6.2.7 Sensor Mechanism	44
6.2.8 Sensor	45
6.3 Manufacturing Milestones	45
7. Testing Plan	45
7.1 Propulsion Testing	45
7.2 System and Subsystem Testing	45

7.2.1 Sensor deployment and recovery mechanism	45
7.3 Manufacturing Testing	46
7.4 Data Collector Testing	47
7.5 Wind Tunnel Testing	47
7.6 Pre-Flight Checklist	48
7.7 Flight Testing Plan	48
8. Performance Results	49
8.1 Propulsion Testing Results	49
8.2 System and Subsystem Testing Results	49
8.2.1 Sensor deployment and recovery mechanism	49
8.3 Manufacturing Testing Results	49
8.4 Wind Tunnel Testing Results	51
8.5 Flight Testing Results	52
9. Bibliography	54

Acronyms and Nomenclature

α , AoA	Alpha or angle of attack	K_v	Motor velocity constant
AIAA	American Institute of Aeronautics and Astronautics	l_{ac}	Distance from aerodynamic center to the center of gravity
AUW	All up weight (Aircraft weight including payload and passengers)	$l_{ac,w}$	Distance from aerodynamic center of the main wing to the center of gravity
AR	Aspect Ratio	$l_{ac,h}$	Distance from aerodynamic center of the horizontal stabilizer to the center of gravity
CA	Cyanoacrylate	LiPo	Lithium polymer
CAD	Computer-aided design	M1	Mission 1
CF	Carbon fiber	M2	Mission 2
CFD	Computational fluid dynamics	M3	Mission 3
CG	Center of gravity	NACA	National Advisory Committee for Aeronautics
CWL	Cubic wing loading	η_h	Dynamic pressure ratio for the horizontal stabilizer
$\bar{c}$	Mean aerodynamic chord	ρ	Density
C_D	Coefficient of drag	PLA	Polylactic acid
C_L	Coefficient of lift	PCB	Printed circuit board
$C_{L,0}$	Coefficient of lift at 0 AoA	Re	Reynolds number
$C_{L,\alpha}$	Coefficient of lift with respect to alpha	RPM	Rotations per minute
$C_{L,\alpha,w}$	Coefficient of lift with respect to alpha on the main wing	S_h	Size of horizontal stabilizer
$C_{L,\alpha,h}$	Coefficient of lift with respect to alpha on the horizontal stabilizer	T	Thrust
$C_{l,\beta}$	Coefficient of roll with respect to beta	T/W	Thrust-to-weight ratio
$C_{n,\beta}$	Directional stability coefficient	UBEC	Universal battery eliminator circuit
$C_{m,\alpha}$	Longitudinal stability coefficient	V_∞	Freestream velocity
d	Diameter	ν	Kinematic viscosity
$\frac{d\epsilon}{d\alpha}$	Washout factor affecting airflow over the horizontal stabilizer with respect to alpha	$V_{logic, min}$	Minimum signal voltage for smooth servo operation
DBF	Design/Build/Fly	$V_{logic, abs min}$	Minimum signal voltage for any servo operation
EPS	Expanded Polystyrene	WU	Washington University in St. Louis
ESC	Electronic speed controller	WUDBF	Washington University Design/Build/Fly
FEA	Finite element analysis		
GM	Ground Mission		
KM Sim	Kinematic mission simulator		

1. Executive Summary

1.1 Overview

This report describes the design, testing, and manufacturing of the *Red Bearon*, Washington University in St. Louis Design/Build/Fly (WUDBF) team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2021 Design-Build-Fly competition. WUDBF aims to complete all missions and maximize expected score in this year's virtual competition. Due to COVID-19 precautions and similar competition requirements to the 2020 Design/Build/Fly competition, such as a high payload capacity and the ability to trail an object behind the aircraft, WUDBF decided a familiar design would be best for both competition scoring and member safety. The *Red Bearon*, therefore, is an updated and revised version of WUDBF's 2020 *Atlas* aircraft and was selected for similar performance reasons, along with manufacturing familiarity for safe construction. Preliminary Design highlights the dynamic stability of the aircraft, resulting in good handling characteristics as proven in flight tests conducted in 2020. The section also shows that two motor configuration optimizes static and dynamic efficiency. Detail Design describes a wood-based structure with a 5 ft wingspan, electronic integration, and an expected Total Mission Score of 5.53. Primary construction materials are balsa wood, basswood, carbon fiber, and UltraCote. Previous testing shows a wood-based aircraft best provides ease of iteration, flexibility, and reliability. Aircraft integration is confirmed in targeted flight tests [1].

1.2 Summary of Design

The *Red Bearon* is a conventional, high-wing, dual motor aircraft with tricycle landing gear, partially optimized for Missions 2 and 3. Derived from a pylon racer-style aircraft, iterative design changes ultimately resulted in a vehicle resembling a military transport aircraft. The *Red Bearon* has a high wing and high tail setup, allowing ample ground clearance for propellers, Fowler flaps, external storage, and internal storage via nose cone opening. The propulsion system is designed for high cruise speeds of 30 m/s and 20 m/s during Missions 2 and 3, respectively, while satisfying takeoff requirements. Designed to carry 4 sensors, the aircraft uses balsa wood construction techniques. Test flights are planned for the near future to provide preliminary competition performance estimates.

1.3 Aircraft Capabilities

Table 1.1 provides basic information about the design of the *Red Bearon* and its projected performance based on aircraft CAD models, a kinematic simulator (KM Sim), and flight test data.

Table 1.1: Basic aircraft information and performance data.

M1 Takeoff Weight [kg]	4.9	Average Cruise Speed [m/s]	25
M2 Takeoff Weight [kg]	5.9	Wingspan [m]	1.524
M3 Takeoff Weight [kg]	5.2	Planform Area [m ²]	0.3618
Top Speed [m/s]	40	Sensor Weight [kg]	0.25
Number of Shipping Containers	4	Estimated Score	5.53

2. Management Summary

2.1 Organization

To facilitate efficient communication and work, WUDBF is organized into an Executive Team, Advisors, and five Design Divisions. Team information and decisions flow downward from the Executive Team to Division Leads and ultimately to the General Members as represented by the hierarchical chart in Fig. 2.1.



Figure 2.1: Team Hierarchy

The Executive Team consists of a Technical President, an Administrative President, a Vice President of Communications, and a Vice President of Finance. Executive Team members are responsible for aircraft technical decisions, administrative tasks, resource allocation, and public relations. Additionally, WUDBF is supported by two categories of advisors: graduate and undergraduate. Graduate Advisors provide industry and/or graduate level expertise and guidance to the team. Undergraduate Advisors are defined as any student who was previously a division lead or on the executive team and they counsel current team leads. There are five divisions (described in Table 2.1) and each is essential to the design and construction of the *Red Bearon*. The number of members assigned to each division is based primarily upon student interest, but also depends on team needs [1].

Table 2.1: Relevant roles, assignments, and skills of WUDBF division personnel. Membership of each division is shown in parentheses.

	Aerodynamics (4)	Structures (4)	Manufacturing (4)	Avionics (6)	SimDev (4)
Roles	- Specify outer mold line of the aircraft - Ensure aircraft meets stability performance goals	- Update airframe - Analyze structure - Manage weight - Create drawings for manufacturing	- Manufacture from CAD - Select materials based on material testing	- Design propulsion systems - Integrate control surfaces - Analyze telemetry	- Develop simulation software - Analyze simulation results
Skills	- Aircraft stability, control, optimization - Numerical/CFD analysis	- CAD - Structural FEA - Mass properties analysis	- Laser cutting, 3D printing, machining - RC aircraft construction	- Propulsion theory - Electrical design - Soldering and crimping	- MATLAB/Simulink - Simulation - Real Flight 8 - Software debugging

2.2 Schedule

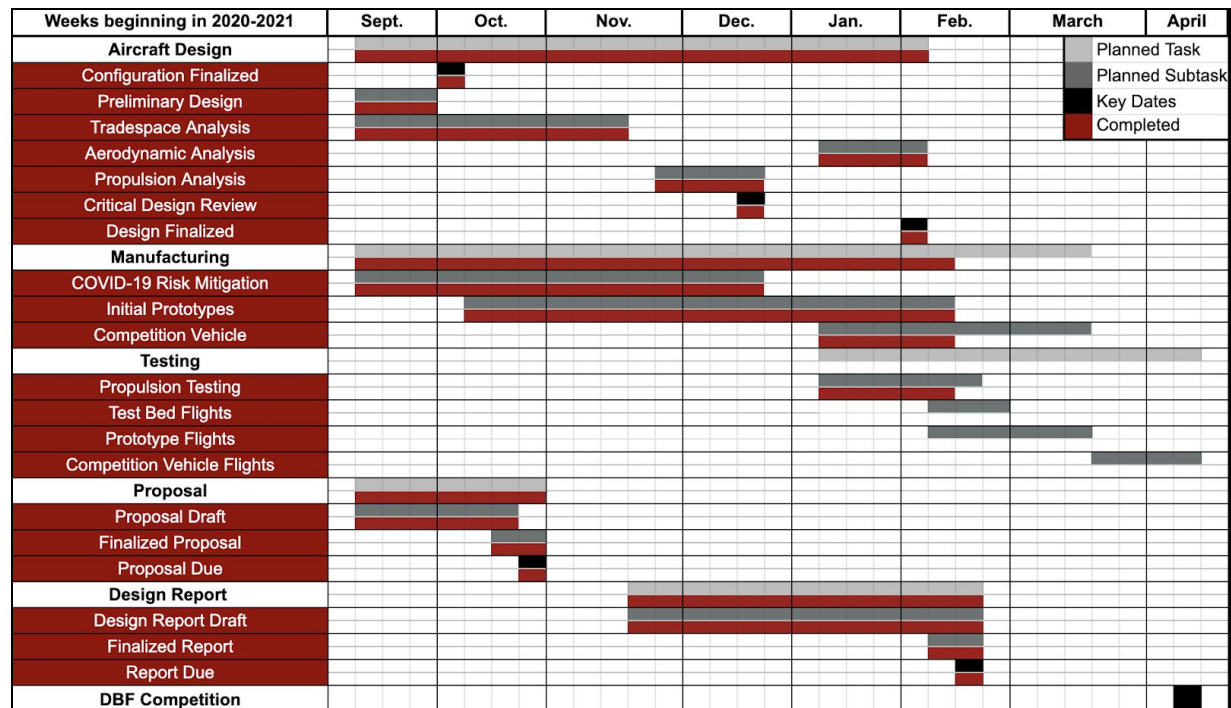


Figure 2.2: Gantt chart of WUDBF's design and build timeline for the *Red Bearon*.

3. Conceptual Design

3.1 Design Philosophy

The conceptual design phase begins with analysis of competition rules, mission requirements, and scoring formulas. The team determines the aspects which most influence the final score of the aircraft, such as lap time, containers carried, sensor length, and sensor weight. The most important capabilities are then translated into design requirements and goals. These goals are used to guide the selection of possible aircraft configurations, which are analyzed by the team. These aircraft are then modeled in XFLR5 to fulfil longitudinal and lateral-directional stability requirements. Finally, the aircraft are simulated for performance and score using WUDBF's in-house mission simulation models. Due to potential adverse weather, WUDBF emphasized risk reduction and aircraft handling qualities to maximize expected score (the analysis factors the probability of configurations successfully completing all missions in combination with predicted scores). The aircraft configurations are compared on predicted scores, perceived risk, and COVID-19 safety during design and manufacturing, guiding final configuration selection. The team selected a Pylon Racer configuration, the same as in 2019-2020, and christened the aircraft the *Red Bearon* [1].

The 2021 DBF competition's mission requirements are based on real-world design challenges. This year, aircraft are required to deploy and retrieve a sensor and carry containers. These capabilities will be tested during four missions, detailed in section 3.2.

3.2 Mission Requirements

Mission 1 (M1), Mission 2 (M2), Mission 3 (M3), and the Ground Mission (GM) will be completed in accordance with the DBF rules and regulations. The team score is determined by Eq. 3.1.

$$\text{Overall Score} = \text{Written Report Score} * \text{Total Mission Score} \quad (3.1)$$

Where *Total Mission Score* is determined by Eq. 3.2

$$\text{Total Mission Score} = M1 + M2 + M3 + GM \quad (3.2)$$

A sample mission path is detailed in Fig. 3.1.

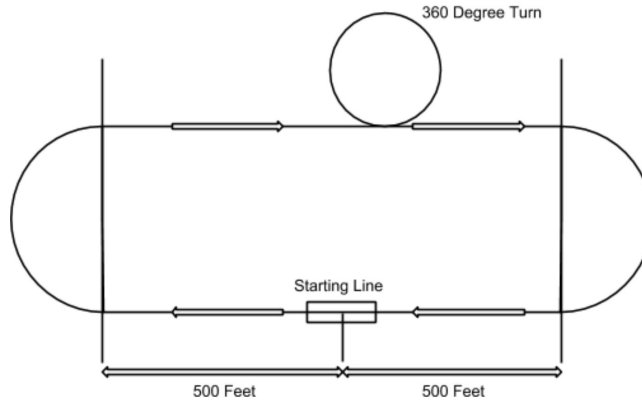


Figure 3.1: AIAA 2021 DBF competition course layout.

3.2.1 Mission 1

M1 is the test flight. The vehicle will begin the mission in its flight configuration without payload. It will take off within 100 ft. Once the aircraft throttle advances, it will have 5 min to complete three laps. The mission is completed with a successful landing. M1 is scored according to Eq. 3.3 [1].

$$M1 = 1 \quad (3.3)$$

3.2.2 Mission 2

M2 is the delivery flight. The vehicle enters the staging box in its flight configuration with payload, which consists of the sensor in a container, its deployment and recovery mechanism, and a number of sensor simulation containers. The vehicle will complete three laps within 5 minutes. After this, the pilot will successfully land the aircraft. M2 is scored according to Eq. 3.4.

$$M2 = 1 + [N_{\#containers/time} / Max_{\#containers/time}] \quad (3.4)$$

Where $N_{\#containers/time}$ is the number of containers carried by the *Red Bearon* divided by its mission time and $Max_{\#containers/time}$ is the highest ratio of number of containers carried to mission time among all teams [1].

3.2.3 Mission 3

M3 is the sensor flight. The vehicle again begins in its flight configuration loaded with the sensor and deployment/recovery mechanism. The vehicle will take off. The sensor will remotely deploy before the first 360 degree turn and be recovered after the last 360 degree turn. The vehicle will have 10 minutes to complete as many laps as possible and then successfully land. M3 is scored according to Eq. 3.5.

$$M3 = 2 + [N_{\#laps * sensor length * sensor weight} / Max_{\#laps * sensor length * sensor weight}] \quad (3.5)$$

Where $N_{\#laps * sensor length * sensor weight}$ is the product of the number of laps flown by the *Red Bearon* and banner length and weight and $Max_{\#laps * sensor length * sensor weight}$ is the highest product of scoring laps flown and sensor length and weight among all teams [1].

3.2.4 Ground Mission

The GM must be completed prior to M2 and M3. The vehicle will begin in a 10 ft square “mission box”, next to the uninstalled sensor in the shipping container. The assembly crew member will enter the box, drop the sensor on all six container sides from 10 inches, remove the sensor from the container to demonstrate no physical damage and all performance requirements for M3. A ground mission official will start a timer. The assembly crew member will load the full M2 payload and leave the mission box, pausing the timer. The pilot will demonstrate flight control functionality. Next, the crew member will return to the aircraft resuming the timer. They will replace the M2 payload with the M3 payload, and leave the box, stopping time. The pilot will prove the flight controls remain active. The plane will be held vertically and sensor deployment and recovery will be demonstrated. The GM is scored according to Eq. 3.6.

$$GM = Min_{time} / N_{time} \quad (3.6)$$

Where N_{time} is WUDBF's time and Min_{time} is the minimum time among all teams [1].

3.3 Design Requirements

Design considerations are centered on meeting M2 and M3 mission requirements. M2 score is maximized primarily by increasing flight speed and container carrying capacity. These requirements call for a design with high dynamic thrust to quickly complete three laps, a large internal volume to store the greatest number of containers, and adequate static thrust and C_L to successfully take off while fully loaded. M3 prioritizes endurance and high flight speed, while also requiring sufficient space to hold the sensor and deployment mechanism. M3 requirements call for an efficient, high aspect ratio (AR) wing with low induced drag and a smaller body to minimize parasitic drag. However, due to wingspan restrictions, higher AR wings limit wing area, decreasing cargo capacity and serving as an AR limit. Optimizing for M3 decreases M2 score as interior storage is reduced. Additionally, larger batteries are required to increase endurance, further decreasing internal storage [1].

Since M2 and M3 prioritize somewhat different parameters, the optimal design balances M2 and M3 requirements. This balances risk and increases expected score. Aerodynamics and Structures must design an aircraft that has high internal volume for container storage, while also balancing low in flight drag (generally correlated with lower operating C_L) and high take off C_L . Avionics must balance propulsion requirements between dynamic and static thrust and aircraft handling. Taking into consideration these various design constraints, WUDBF designed several prototypes for further evaluation [1].

3.4 Initial Aircraft Designs

After preliminary analysis was completed for the 2020 competition, design requirements yielded six aircraft designs that could meet the mission requirements. Due to the similarities in the 2020 and 2021 competition requirements, these designs were each considered again for the 2021 competition. Each aircraft was designed in XFLR5 and examined a different geometry, propulsion, avionics, and container count. All six aircraft met basic aircraft stability requirements. The conventional configurations used NACA 2412 airfoils and the deltas NACA 0010 for their realistic $C_{L,0}$ and well defined aerodynamic characteristics. These aircraft are presented in Fig. 3.2 [1].

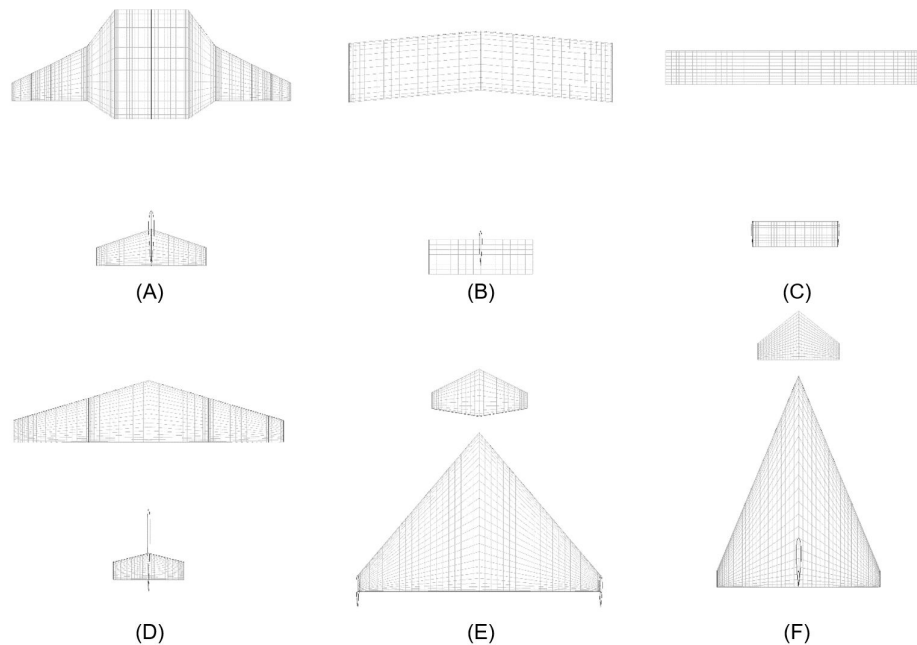


Figure 3.2: Initial aircraft planforms. A: Blended Body, B: Cargo Aircraft, C: Twin Boom, D: Pylon Racer, E: Delta Canard 5 Ft, and F: Delta Canard 3 Ft [1].

3.4.1 Blended Body

The Blended Body is a blended wing body, conventional-tail configuration that began as a flying wing utilizing a reflexed airfoil. The benefits of the flying wing were outweighed by the complexity of the design, so an empennage was added while preserving the blended, airfoil-shaped body. This design offers flexibility between tractor and pusher motors, and between single and twin motor configurations. Significant portions of the wing can be flapped for low-speed flight and the interior volume of the body provides a high degree of flexibility for payload size and configuration. The main drawback of the design is manufacturing difficulties, specifically with regard to stiffness in the “body” section [1].

3.4.2 Cargo Aircraft

The Cargo Aircraft is a high-wing, conventional-tail configuration that sacrifices takeoff and cruise performance for maximizing payloads. It is designed to excel in M2, holding up to 4 heavy sensors, at the cost of M3 score. The Cargo Aircraft’s wing is large to maximize lift and swept back to increase

longitudinal stability and place the center of gravity (CG) over the main payload compartment. This entails a large fuselage volume for sensors. The Cargo Aircraft is somewhat flexible in motor configuration — it can support both a single and twin tractor motors. High lift device integration is simpler on a conventional wing [1].

3.4.3 Twin Boom

The Twin Boom is a high-wing, H-tail configuration that was fully designed as the 2019 competition vehicle and updated for this year. It highlights the benefits of having multiple fuselages at the costs of additional parasitic drag. This particular twin-boom aircraft has a twin puller motor configuration and does not employ the thrust-vectoring seen in WUDBF's 2019 aircraft. Instead, it only uses blown trailing edge flaps for low speed flight. Having two separate fuselages presents manufacturing difficulties of rigidity and weight distribution that are otherwise nonissues [1].

3.4.4 Pylon Racer

The Pylon Racer is a low-wing, T-tail aircraft that was fully designed as last year's competition vehicle and updated for this year based on the class of aircraft by the same name built for high speed maneuverability. These aircraft benefit from lower wing loading, which decreases maximum payload. It has a single tractor motor configuration and a high degree of flexibility for high lift devices. Pylon racing aircraft are not designed for high payload capacities, requiring increased wing area and wing loading. One key concern of this is maneuverability, necessitating additional analysis [1].

3.4.5 Delta Canard 5 Ft

The Delta Canard 5 Ft is a delta-wing canard, twin-tail configuration with a 5 ft wingspan. This necessitates a large wing area, with rigidity becoming a major manufacturing concern. Despite this, the configuration is favorable due to high maneuverability at high angles of attack (AoA). Single or twin motors must be pushers to not disturb the delta airflow. However, this eliminates the possibility of blown flaps which, in conjunction with the delta wing's poor takeoff performance, represents a major drawback for takeoff weight. This is the main subject of further analysis for this configuration [1].

3.4.6 Delta Canard 3 Ft

The Delta Canard 3 Ft is a delta-wing canard, conventional-tail configuration which accounts for the rigidity concerns of the Delta Canard 5 Ft and is specialized for high-speed, low-capacity flight. This is the only aircraft that has a wingspan of less than 5 ft. The 3 ft wingspan ensures manufacturing concerns are manageable. Performance concerns are not, however, solved by this model, necessitating a lower payload capacity. This aircraft's performance determines whether a low capacity, high speed configuration is preferable over a high capacity, low speed configuration. The motor configuration is unchanged from the Delta Canard 5 Ft [1].

3.4.7 Configuration Qualitative Comparison

Table 3.1 presents a summary of the qualitative differences between configurations. A “+” implies advantages, a “-” disadvantages, and “=” means that the compared categories are similar.

Table 3.1: Qualitative differences between considered aircraft configurations [1].

Category	Blended Body	Cargo Aircraft	Twin Boom	Pylon Racer	Delta Canard 5 Ft	Delta Canard 3 Ft
M2 Performance	=	+	=	=	-	-
M3 Performance	+	-	=	=	+	+
Manufacturability*	=	=	=	+	=	=
Propulsion	+	=	+	-	-	-

*Due to COVID-19, the Pylon Racer has the highest advantage as a similar aircraft has been constructed previously.

3.5 Final Design Selection

3.5.1 Initial Configuration Scoring Analysis

To guide team design decisions, each aircraft flew simulated missions in KM Sim to produce performance and mission scoring estimates. KM Sim is a MATLAB-based idealized control dynamics simulation of an aircraft using a prescribed course. Aircraft aerodynamic properties, inertial properties, and mission-specific parameters were manually compiled for each airplane. The simulator uses configuration specific data to model relevant states during each mission. The aircraft aerodynamics and propulsion models are table-lookup values from XFLR5 and the University of Illinois at Urbana-Champaign propeller database, respectively [1].

Simulated flight phases in KM Sim include ground roll, rotation, optimum climb, max climb, cruise, constant radius and constant g-load turns, and optimum glide. Each flight phase is modeled as a separate system of differential equations. It is numerically integrated, producing a time history of 14 states, including velocity, orientation, battery charge, and other useful data. This time history is used for data visualization and scoring. A sample trajectory plot, shown in Fig. 3.3, displays the complete flight path of an aircraft using position and flight phase data. Flight paths are plotted within an East-North-Up coordinate space [1].

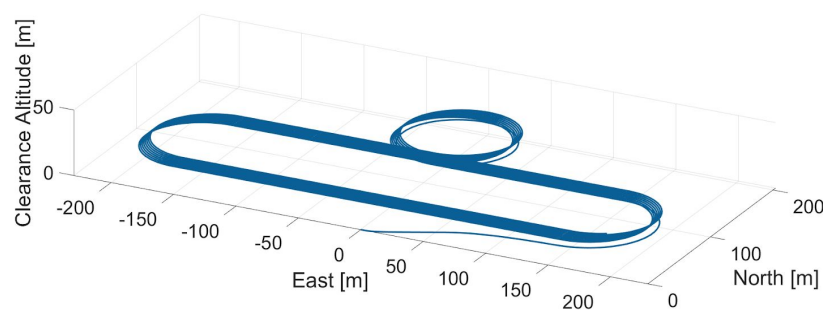


Figure 3.3: Trajectory plot of KM Sim results [1].

KM Sim also produces graphs of important states, shown in Fig. 3.4, as an eight-panel results summary of the flight. The flight summary contains time history plots of the most useful states. These data are useful for quick visual analysis to make sure the simulated flight was performing as required. Some examples include checking maximum and nominal values of AoA, the mission phase in case of a failed flight, and remaining battery charge [1].

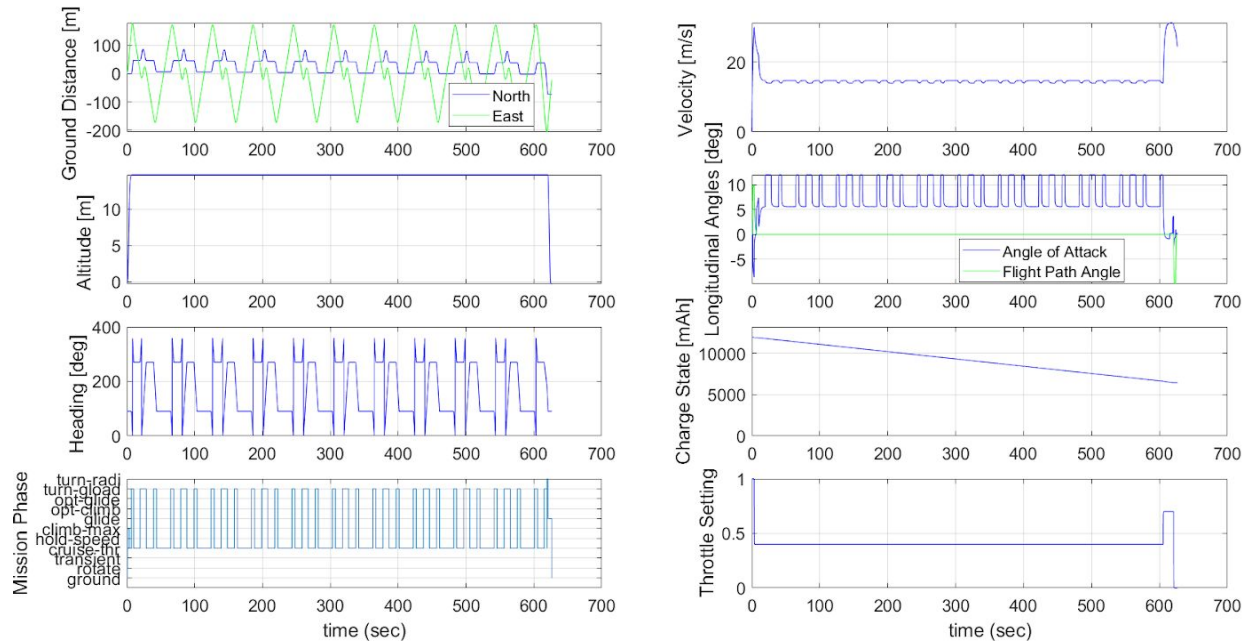


Figure 3.4: Example of eight panel results summary [1].

KM Sim has the necessary features to simulate each of this year's missions. For example, KM Sim alerts the user when the aircraft did not takeoff within 100 ft, but continues the simulation for analysis of the rest of the flight. It also supports aircraft configuration changes, like flaps, and loadout changes, such as unfurling the banner. Configuration and loadout changes are performed by switching or modifying the aerodynamic lookup tables. KM Sim can be employed for a wide variety of studies. For instance, aircraft designs can be run through different courses with the same initial state or through the same course with different payload, flap, and thrust vectoring instructions. Additionally, parameter studies, range calculations, and maximum takeoff weight calculations are all easily performed in KM Sim [1].

3.5.2 Sensitivity Analysis

Due to competition limitations on wingspan, the aircraft AR is strongly correlated to the planform area of the aircraft. The additional efficiency of a high AR wing must be balanced with the additional lift of a larger, low AR wing. The high payload requirement of the competition slightly favors a larger wing area at the cost of efficiency and the sensitivity study in Fig. 3.5 demonstrates this; the Blended Body performs best due to its maximized main wing area. A partial derivative value of -0.043 score/AR was found in the analysis and the final *Red Bearon* design had an AR of 6.4 [1].

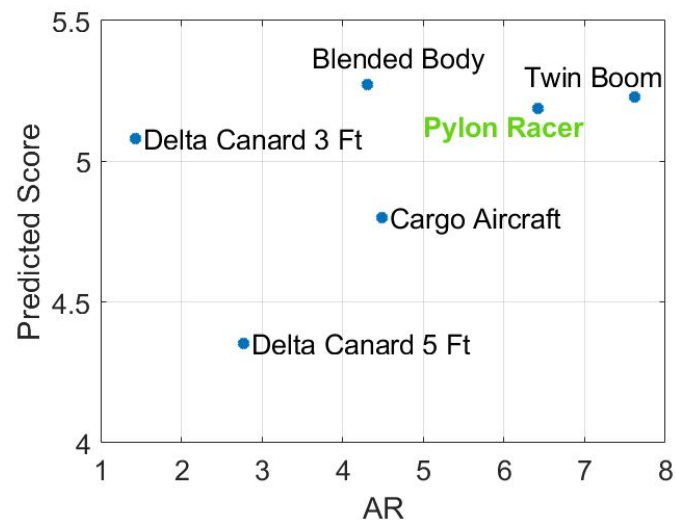


Figure 3.5: Scoring plot of estimated total score and aspect ratio [1].

The second sensitivity study entails battery mass; a larger mass increases either power or endurance but impairs wing loading and takeoff distance (see section 4.4.1). The battery mass partial is found via the chain rule from partials with respect to voltage and charge capacity, and reveals that increasing battery mass has a positive (albeit small) impact on score. Since the chosen battery mass lies at the upper limit of allowed energy (and thus mass), the battery optimizes score for this configuration. The analysis yielded a value of 0.003 score/kg for the battery mass.

3.5.3 Aircraft Risk Analysis

To execute the team's strategy of maximizing the most probable score, aircraft design risk must be reduced. WUDBF rated each of the aircraft on five risk-related categories to quantify each design's overall risk. The CAD score represents the difficulty to create the aircraft within SolidWorks. The Make score captures the complexity of sourcing raw materials and creating the aircraft components. Assembly reflects the difficulty of assembling all of the aircraft's components, especially under COVID-19 restrictions. Structure refers to the difficulty of making the configuration structurally sound. Finally, Stability embodies the difficulty of ensuring longitudinal and lateral stability as well as favorable handling qualities. This risk matrix is shown in Table 3.2 and then plotted with expected score in Fig. 3.5. Note that the two Delta Canard configurations were given the same risk scores due to their similarity [1].

Risk categories were given weights to reflect their severity, with higher weights reflecting more serious risk. The CAD risk category was weighted at 5% because the 2019-2020 CAD model was used for the *Red Bearon*. The Make category was given a weight of 30% because component sourcing and fabrication may be out of the team's control and due to a limited number of people able to manufacture. A design that requires expensive materials and components that require specialized tools represents risk that cannot be overcome as easily as risk associated with other categories. Assembly risk was weighted at 35% due to the team's historical struggles with aircraft manufacturing and COVID-19 restrictions.

Structure and Stability were each weighted at 15% because they represented the viability of the configuration itself and were not dependent on the team.

Table 3.2: Aircraft configuration risk matrix.

Risk Category	Weight (%)	Cargo		Blended Body		Mission 3 Delta		Pylon T Tail	
		Rating	Weighted	Rating	Weighted	Rating	Weighted	Rating	Weighted
CAD	5	1.1	0.055	3.5	0.175	3	0.15	1	0.05
Make	30	1.5	0.45	3	0.9	4	1.2	1	0.3
Assembly	35	1	0.35	2	0.7	4	1.4	1	0.35
Structure	15	1	0.15	3	0.45	4	0.6	1	0.15
Stability	15	1	0.15	2.8	0.42	2.5	0.375	1.1	0.165
Total score		1.155		2.645		3.725		1.015	
Rank		2		3		4		1	

The Cargo Aircraft and Pylon Racer have low risk scores due to a simpler wing and fuselage design. Further, their smaller wings simplify manufacturing. They can also easily be made strong and stable. The Blended Body has high risk scores due to the anticipated complexity of the airframe, the more complicated components, and the aerodynamic characteristics simulated by the Aerodynamics Division. The Delta Canards have extremely high risk scores because of the large size of the wing. The team estimated that it would be costly and difficult to source wood and other components large enough for the wing ribs and furthermore did not have tools large enough. As a result, the Manufacturing Division determined that the Delta Canard would be nearly impossible to make. The Twin Boom configuration has a medium-high risk score due to anticipated structural challenges in making the plane rigid given the high moment on the booms connecting the tail to the fuselage. These rigidity issues also represent stability risks, resulting in a medium-high stability risk score [1].

Each aircraft is plotted for its normalized M2, M3, and weighted risk, shown in Fig. 3.6. The score is calculated assuming that each aircraft flies all three missions, where only M2 and M3 are displayed as performance in M2 and M3 determines what additional score [1].

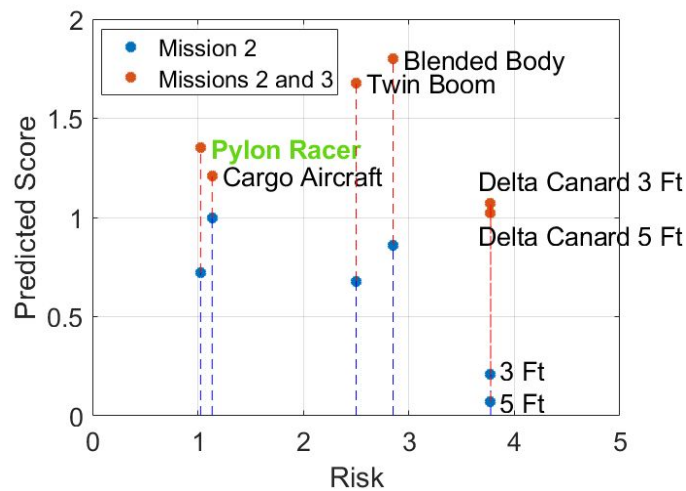


Figure 3.6: Scoring plot of estimated M2, M3, and measured risk [1].

Several aircraft converged to a final score within a half of a point of one another at the top of the graph. WUDBF interpreted this result as a demonstration of score independence relative to configuration. As a result, the configuration choice could be made based on factors beyond performance, such as manufacturability [1].

3.5.4 Final Configuration Decision

The final aircraft configuration of the *Red Bearon* was informed by the analyses of the six initial designs. The *Red Bearon* most closely resembles the Pylon Racer configuration due to its high predicted score and minimal associated risk. The *Red Bearon* configuration has a high-wing with a swept leading edge, a straight trailing edge, and a conventional tail. The straight trailing edge ensures that the high lift devices do not need to be mounted at an angle, reducing aerodynamic concerns of spanwise flow. The leading edge is swept to achieve a taper ratio of 0.4 to minimize induced drag. The high-wing eliminates the need for dihedral and provides greater ground clearance for the trailing edge high lift devices and propellers. The conventional tail was chosen to minimize the necessary structure in the vertical stabilizer [1].

4. Preliminary Design

4.1 Aerodynamic Performance

The *Red Bearon* design was iteratively improved to ensure the plane is stable, controllable, and achieves maximum aerodynamic performance. Both XFLR5 and MATLAB were used for aircraft analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were dimensioned to improve handling characteristics and minimize takeoff distance, respectively [1].

4.1.1 Main Wing Planform Area

To maximize lifting capacity, the wingspan is 5 ft, the maximum allowed. A manufacturing constraint was a straight trailing edge for the trailing edge Fowler flaps. Finally, the performance goals were to cruise at 30 m/s near 0° AoA while having a stall speed of 8 m/s with flaps extended. Given these constraints and a desired taper ratio of 0.4, a wing area of 0.36 m^2 was selected, yielding 58 N of lift at cruise, a cubic wing loading (CWL) factor of 21 oz/ft³, and an AR of 6.4 [1].

4.1.2 Wing Design for Aerodynamic Performance

Span limited aircraft with an elliptical lift distribution have minimal induced drag; this distribution is common in most conventional aircraft [2, 3]. For the main wing, the Aerodynamics Division applied twist and tapering to approximate an elliptical lift distribution. Along the span, the twist was applied at four locations, shown in Table 4.1. Minimizing these locations aids in manufacturing. The actual lift distribution closely models the ideal elliptical shape, shown in Fig. 4.1. Additionally, by de-twisting the wing at the wingtips, the aircraft maintains roll control in a partial stall and tip-stall tendencies are weakened [1].

Table 4.1: Main wing twist values as a function of distance from the root [1].

Distance from root [mm]	Twist [$^\circ$]
0	0
339	0.5
677	0
762	-4.0

Figure 4.1: Elliptical ideal [black] and actual [red] lift distribution of main wing at 0.74° AoA [1].

4.1.3 Main Wing Airfoil Selection

The Reynolds number (Re) significantly reduced the scope of the main wing airfoil selection. The predicted Re of the plane was calculated, using Eq. 4.1.

$$(4.1)$$

where $\bar{c}$ is the mean aerodynamic chord [m], V_∞ is the freestream velocity [m/s], and ν is the kinematic viscosity of air [m²/s]. Using XFLR5, kinematic viscosity was calculated from the average temperature in April (14 °C) and altitude (400 m) of Wichita. As the flights for the 2020-2021 aircraft will take place in St. Louis, the team decided to reuse these parameters as they will be almost the same as Wichita. The team searched the University of Illinois at Urbana-Champaign database for airfoils with a maximum C_L/C_D ratio

at the cruise Re at AoA between 0° and 2° which also performed well at low Re and high AoA. In the search parameters, the maximum thickness was set to 12% to use thin airfoil theory approximations. Additionally, a minimum thickness was set to 8% for structure and system integration. The maximum camber was set to 4% to minimize drag while maintaining high lift. The results returned three potential airfoils as shown in Table 4.2 [1].

Table 4.2: Final airfoil selection [1].

	Max. C_L/C_D at Re = 500 000	Max. Thickness [% of chord]	Max. Camber [% of chord]
Eppler E193	114	10.2	3.0
Eppler E195	112	11.8	2.7
Clark Y	98.7	11.7	3.4

The Eppler E193 was chosen since it has the highest C_L/C_D at the cruise Re and performs well at low Re and high AoA, all while maintaining a high level of manufacturability. Additionally, WUDBF has experience with Eppler airfoils. Figure 4.2 shows the Eppler E193 airfoil.

Figure 4.2: A cross section of the chosen Eppler E193 airfoil [1].

4.1.4 Stabilizer Airfoil Selection

Stabilizer airfoils are generally symmetric as they may either provide upforce or downforce, depending on the final trim state of the aircraft. The team selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects [1].

4.1.5 High Lift Devices

WUDBF employs a series of high lift devices to generate enough lift on takeoff. Additionally, WUDBF has knowledge on Fowler flaps, which were implemented to decrease takeoff length in last year's aircraft. The theoretical additional lift values for the Fowler flaps are from published data [4]. The Fowler flaps are placed in the propwash of the motors, increasing the relative freestream air speed over the flaps, increasing their effectiveness in low-speed, high-power flight. The flap has a chord fraction of 0.3, a span of 350 mm, and deploys at a 39° angle. The increment of section maximum lift coefficient at 39° -deflection is $\Delta C_{L_{max}} = 1.75$ for the Fowler flap, which is greater than either of the other slotted flap options it is compared against [4]. A "flap fusion" mode was also implemented where the ailerons deploy downward as well, increasing the total deployed area of the wing [1].

Leading edge high lift devices were considered because they further decrease takeoff speed by increasing maximum AoA and thus, the maximum C_L . Specifically, Krueger flaps offer the best compromise between high lift performance and weight. Krueger flaps were designed with published guidelines [5]. The design parameters were a rotation angle of 130° , a flap length of 20% of the chord,

and a flap nose thickness of 5 mm [5]. They were ultimately removed to streamline manufacturing and maintain a structural leading edge [1].

Winglets will be implemented in the next iteration to increase lift performance and were tested at various speeds in viscous and inviscid analyses using XFLR5. The additional lift generated as a function of winglet area hits a plateau beyond a certain point; the area was therefore chosen as the point at which doubling the winglet area only led to a 10% increase in additional lift: 94 cm² per winglet. The winglets are upward facing for ground clearance. The Manufacturing and Structures Divisions determined that this design is feasible and the additional lift generated outweighs increased manufacturing complexity.

4.1.6 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using basic moment equations and rule-of-thumb parameters proven on past aircraft. The moment balance equations are Eq. 4.2 through Eq. 4.4

$$L_p p + L_{control} = 0 \quad (4.2)$$

$$M_q q + M_{control} = 0 \quad (4.3)$$

$$N_r r + N_{control} = 0 \quad (4.4)$$

The lateral and longitudinal derivatives were taken from an XFLR5 dynamic stability analysis. The “control” moments were also generated in XFLR5 using deflections of the control surfaces at varying speeds. The control surface sizes on the model were tweaked until the rates met the pilot’s standards. Chord fractions for the control surfaces on the first prototype were 40% for the ailerons, 30% for the elevator, and 20% for the rudder. These control surface sizes were validated in flight. The pilot listed the initial aircraft as a 3 on the Cooper-Harper scale, indicating satisfactory but suboptimal flight characteristics. The aircraft was slightly too responsive in roll while greater pitch and yaw authority would increase pilot comfort. The rudder and elevator were enlarged on the next iteration in response.

4.2 Model Capabilities and Uncertainties

KM Sim is a simulator of ideal aircraft performance that is useful for initial configuration choice and basic mission performance modeling, but cannot be used as a high-fidelity simulator. The flight path is preplanned and does not take into account wind gusts or pilot control. The initial aerodynamic calculations are based on table look-ups of pre-calculated data from XFLR5, which poorly models parasitic drag. However, KM Sim can take in data from any source in table format, allowing for improved modelling from wind tunnel data (see section 8.4). Aircraft surface areas are modeled as flat plates to estimate parasitic drag. KM Sim provides an estimate of flight performance metrics (Fig. 4.3). The results allow the team to remove unviable configurations, making final selections easier [1].

After configuration selection, KM Sim was upgraded with empirical data from flight tests, motor tests, and wind tunnel tests. Flight test data provided the required throttle setting to fly at cruise speed, which helped correct inaccurate drag calculations. Motor test stand data provided the correct maximum

thrust value and the throttle breakpoints values. Finally, wind tunnel test data provided more accurate C_L data due to features not modeled in XFLR5. With these additional data sources, KM Sim outputs closely matched flight test data (see section 8.5), shown in Fig. 4.3. The left plot represents the straight section of flight chosen from the flight test, and the right plot represents the simulated flight using the updated KM Sim. From visual inspection during the test flight conducted in 2020, the *Atlas* flew at an AoA range of 0-2°, confirmed by KM Sim. Due to the airframe remaining unchanged for the 2020-2021 competition, it is expected the *Red Bearon* would fly similarly [1].

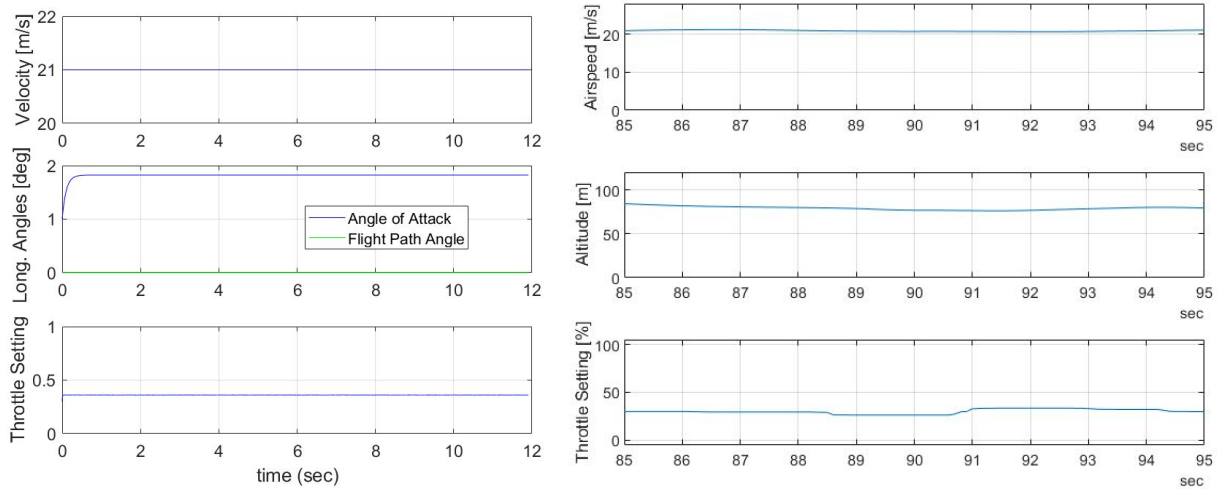


Figure 4.3: Left: KM Sim predicted performance. Right: flight test data from 2020 [1].

4.3 Aerodynamic Analysis

4.3.1 Longitudinal Stability and Horizontal Stabilizer Sizing

Longitudinal stability is fundamental to basic aircraft design and is a function of the location of the center of mass, the added lift provided by the main wing, and the added lift provided by the horizontal stabilizer. The center of mass is liable to shift between missions with the addition of passengers and cargo or a banner. Ensuring that the aircraft's longitudinal stability falls within the desired range for all loading configurations is vital for a successful mission and competition. Longitudinal stability is dependent on several factors which are shown in Eq. 4.5 [1, 5].

$$C_{m_\alpha} \equiv \frac{\partial}{\partial \alpha} C_m = C_{L_{\alpha w}} \frac{l_{ac_w}}{\bar{c}} + C_{L_{\alpha h}} \left(1 + \frac{\partial \varepsilon}{\partial \alpha} \right) \eta_h \frac{S_h}{S} \frac{l_{ac_h}}{\bar{c}} \quad (4.5)$$

Equation 4.5 shows the distance from the aerodynamic centers of the main wing and horizontal stabilizer to the CG ($l_{ac,w}$, $l_{ac,h}$), and the size of the horizontal stabilizer S_h are variables that affect longitudinal stability. Elevator distance was driven by longitudinal eigenvalue analysis, seen in section 4.3.3. Based on prior aircraft designs of similar size and team experience, $C_{m,\alpha}$ values between -0.5 and -1.8 were appropriate. The $-1.05 C_{m,\alpha}$ at $C_m = 0$ fit within the bounds of the guidelines and is resilient to changes in configuration shifting the center of mass [1].

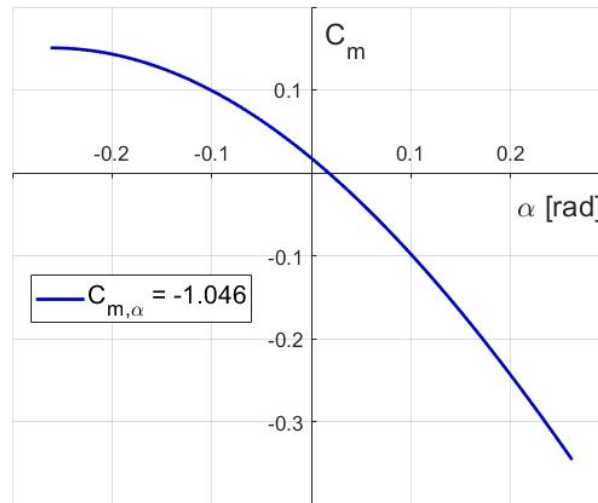


Figure 4.4: Longitudinal stability curve, C_m vs. AoA (α) [1].

4.3.2 Directional and Lateral Static Stability

Directional and lateral stability modes were determined after completing longitudinal stability analysis. Since the lateral directional stability space links lateral and directional modes, individually looking at their static stability derivatives does not provide a complete view of the aircraft dynamics. An analysis of the dynamic stability poles was deemed more useful for understanding directional and lateral stability. Figure 4.5 shows the lateral and directional static stability slopes which are as follows: roll stability is negative with a $C_{l,\beta}$ of $-1.29\text{e-}3 \text{ rad}^{-1}$ while the directional stability, $C_{n,\beta}$, is $3.06\text{e-}3 \text{ rad}^{-1}$.

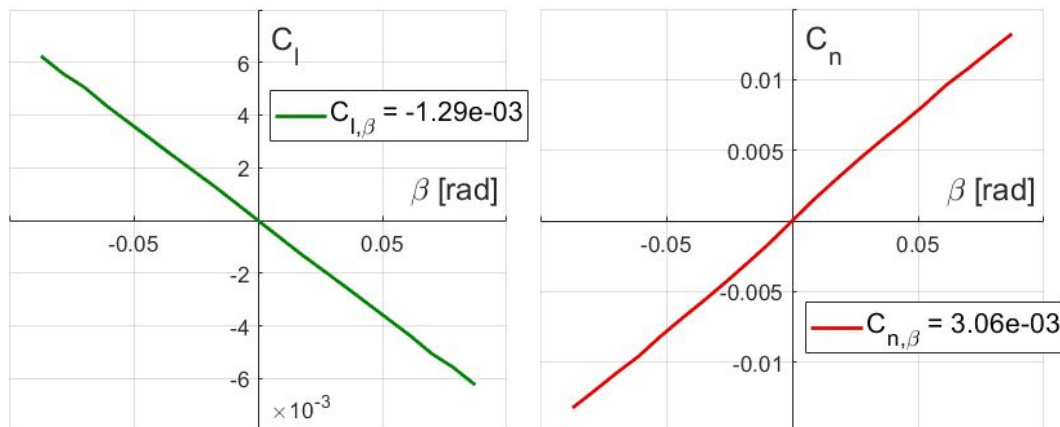


Figure 4.5: Left: Lateral stability curve, C_l vs. β . Right: Directional stability curve, C_n vs. β [1].

4.3.3 Longitudinal Eigenvalue Analysis

Longitudinal eigenvalue analyses conducted in XFLR5, shown in Fig. 4.6, drove horizontal stabilizer distance from the CG. When plotted on the real and imaginary axes, the *Red Bearon's* complex roots for the longitudinal stability case lie in the left (negative) half plane, which is necessary to ensure stability [6]. WUDBF chose to constrain analysis and iteration to the short period mode since the long period mode has a high enough time scale such that minor pilot control inputs can eliminate oscillations [6]. Iterating through a range of tail distances, WUDBF determined the proper tail distance by comparing

the damped and undamped frequencies to those of other aircraft, shown in Fig. 4.7. The final shape of the longitudinal eigenvalues, is similar to other aircraft and serves as additional confirmation that aircraft dynamics have been predicted correctly [1, 6].

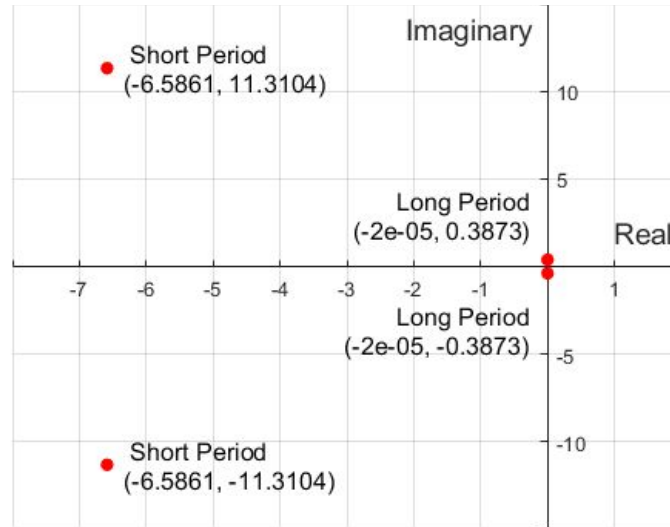


Figure 4.6: Longitudinal eigenvalues characteristic short and long period modes for final horizontal stabilizer [1].

Table 4.3 displays the short period and long period natural and damped frequencies obtained from the eigenvalue analysis in Fig. 4.6.

Table 4.3: Short period and long period natural and damped frequencies [1].

Mode	Natural Frequency [Hz]	Damped Frequency [Hz]	Damping Ratio
Short Period	2.08	1.80	0.503
Long Period	0.062	0.062	0.00051

The *Red Bearon*'s natural frequency and damping ratio of the short period mode were found to provide acceptable handling qualities at cruise using XFLR5's dynamic stability analysis tools. The damping ratio was satisfactory and the natural frequency, while large for full-size aircraft, is in line with other small UAVs [7]. The *Red Bearon*'s short period mode natural frequency plotted versus wingspan follows the trend of a higher natural short period frequency for a smaller aircraft, shown in Fig. 4.7 [1, 7].

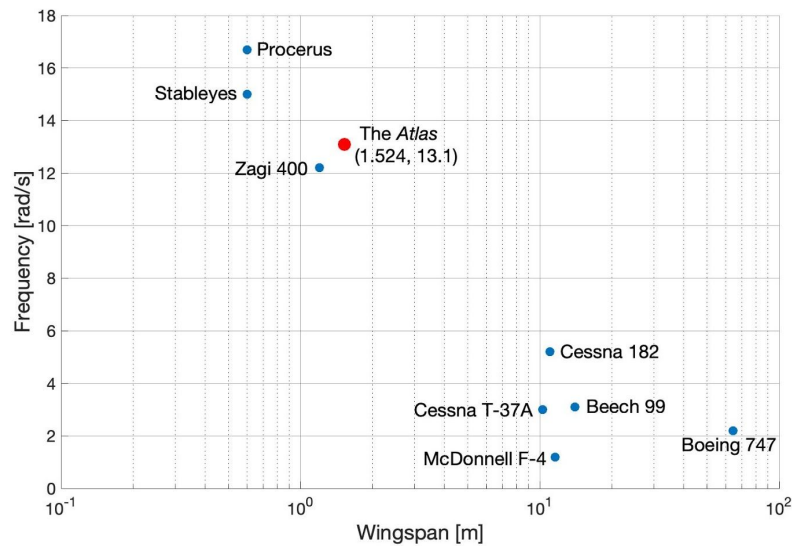


Figure 4.7: Natural short period frequency vs. wingspan for various aircraft [1, 7].

4.3.4 Lateral-Directional Eigenvalue Analysis

The lateral-directional eigenvalue analysis determined vertical stabilizer size and position. The Dutch roll stability characteristics were compared to other aircraft in Fig. 4.9 to ensure the *Red Bearon's* eigenvalues were in line with similar aircraft. As shown in Fig. 4.8 and Table 4.4, the spiral mode and roll mode time-to-halve are extremely large and small, respectively, which was expected. Having a high time-to-halve for spiral mode and low time-to-halve for roll mode are characteristics of a laterally stable aircraft, therefore no design changes were made based on these two roots [6]. The *Red Bearon's* lateral-directional eigenvalues achieve dynamic stability in addition to satisfactory handling qualities [1].

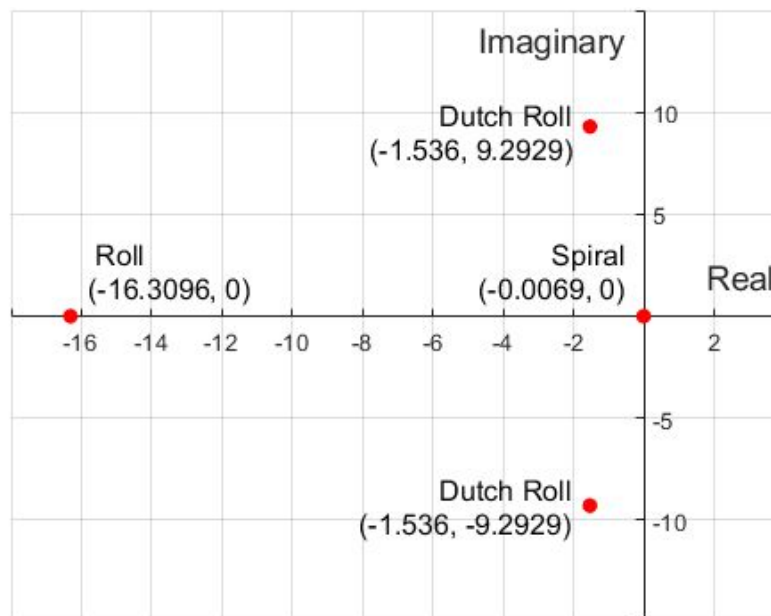


Figure 4.8 Lateral directional eigenvalues: roll root, spiral root, and Dutch root [1].

Table 4.4 displays the characteristics of the Dutch roll eigenvalue.

Table 4.4: Dutch roll, Spiral, and Roll Modes [1].

Mode	Natural Frequency [Hz]	Damped Frequency [Hz]	Damping Ratio	Time to Halve [s]
Dutch Roll	1.06	1.05	0.141	-
Roll	-	-	-	100.1
Spiral	-	-	-	0.042

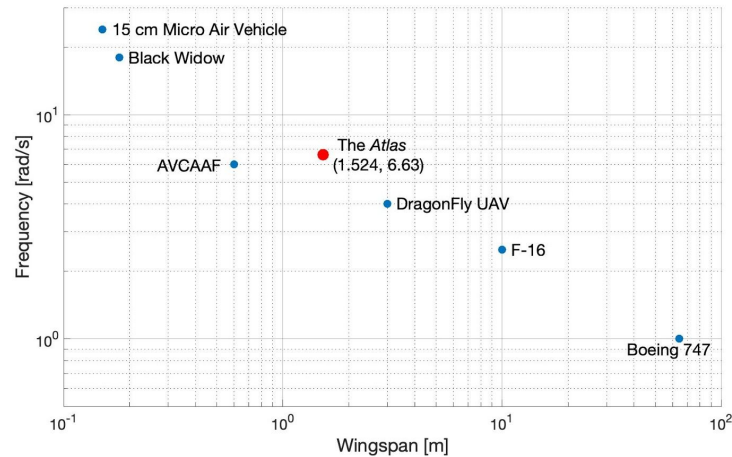


Figure 4.9: Dutch roll natural frequency vs. wingspan size overlaid with other aircraft [1, 7].

Lateral stability poles for previous design iterations are shown in Fig. 4.10 as part of a sensitivity study of empennage sizing. The “Final” configuration is on the log-log trend of Fig. 4.9.

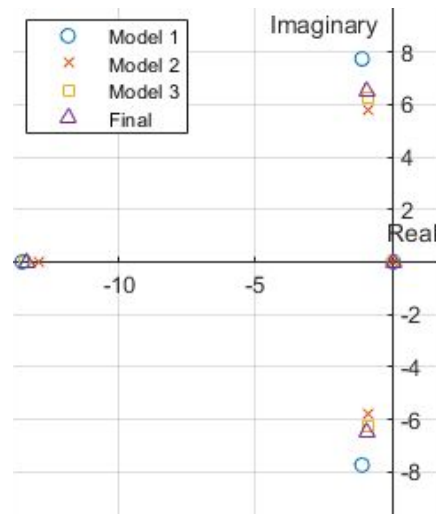


Figure 4.10: Lateral stability root locus plot with early iterations graphed [1].

4.3.5 Drag Determination

While XFLR5 provided accurate lift and induced drag measurements, the analysis considered only inviscid flow. Parasitic drag and drag due to body features were not included in the model. A parasitic

drag script was created to compensate, modeling objects as thin plates. The thin-plate drag estimate was then corrected for thickness using wind tunnel data correlations (See Section 8.5) [1].

4.3.6 Sensor Analysis

Due COVID-19 restrictions, testing of the Sensor mechanism has not been completed at the time of writing this report. Sensor testing will be completed over the next month, using a convertible for bench tests. Once the aircraft is complete, flight test data will be collected as well.

4.4 Propulsion Sizing

The propulsion system for the *Red Bearon* was designed to maximize expected total mission score by maximizing flight speed while adhering to aerodynamic, structural, and takeoff requirements. Based on analysis of M2 and M3 requirements, a balance of high flight speed, adequate static thrust, and reduced risk are the most important considerations for maximizing the expected score. The goal of the propulsion system is therefore to maximize dynamic thrust, which is directly related to flight speed, while keeping in mind three key constraints:

1. Handling Characteristics: Low wing and power loading to provide good handling qualities to decrease risk.
2. Endurance: High cruising efficiency to last the full 10-minute time window for M3.
3. Static Thrust: Sufficient thrust-to-weight ratio to carry the maximum payload for M2.

4.4.1 Initial Sizing

Preliminary propulsion sizing was based initially on cubic wing loading (CWL) a parameter used for grouping RC planes based on flight characteristics. CWL is given by the following equation:

$$CWL = \frac{AUW}{A^{1.5}} \quad (4.6)$$

Where AUW is all up weight and A is wing planform area. Based on prior experience, the *Red Bearon* ideally should handle like a mix of a scale plane and a racer for M3, the primary performance and endurance mission. Data collected from various model aircraft suggest that these types of planes have a maximum CWL of 20 kg/m³ [8]. The initial design of the *Red Bearon* therefore aimed to have a high CWL of 20 kg/m³. AUW was then derived from this wing loading and a wing planform area of 0.36 m² using the equation shown above. This sets the desired AUW at 4.5 kg for M3.

General guidelines suggest that for a plane seeking high performance characteristics and acrobatic ability like the *Red Bearon*, the power loading should range from 265 - 397 W/kg (120 - 180 W/lb) [9]. The upper end of this range, 397 W/kg, was targeted in order to balance the high CWL of 20 kg/m³ chosen previously. Based on the desired AUW of 4.5 kg, and assuming a typical efficiency of 80%, this sets the desired power output at ~2250 W.

4.4.2 Motor Configuration

The *Red Bearon* can support both one- and two-motor configurations (see section 3.4.2), but analysis favors two. The two motor configuration optimizes motor-propeller efficiency by increasing total propeller disk area without the need for an overly large propeller. In addition, the two motor configuration alleviates ground clearance issues and simplifies container loading. One drawback of a multi-motor configuration is the additional weight of a second motor. However, a survey of all brushless electric motors found on the popular website Hobbyking reveals that motor mass correlates linearly with power rating, so that two motors weigh about the same as one for the same total electrical power [1].

4.4.3 Battery Specifications

LiPo batteries were chosen over NiMH because of their higher energy and power densities. To maximize electrical efficiency and minimize electrical losses from parasitic resistance, high voltage 6S packs were selected. To stay within competition requirements batteries can at most hold 100 Wh each. The 6S voltage choice therefore sets the maximum possible charge capacity at 4500 mAh. The overall energy limit of 200 Wh means that at most two 6s 4500 mAh LiPo batteries can be used.

4.4.4 Propulsion System Selection

Using a custom propulsion simulation in addition to online sizing tool eCalc, multiple propulsion systems were evaluated based on their maximum flight speed and also their ability to deliver the desired 2250 W of power, fly a full 10 minute M3, and provide sufficient static thrust to allow for large payloads for M2. Based on this analysis, two Sunnysky X4125 V3 420 K_v motors each swinging a 15x10 propeller were found to meet all performance requirements. In order to fly the entire 10 minute M3 (allowing for a small factor-of-safety), the full 200 Wh was required leading to the selection for the battery pack of two Tattu 4500 mAh 6S, 25C LiPo batteries connected in parallel. Table 4.5 shows the selected preliminary propulsion package. The results of the propulsion simulation for the selected package is shown in Table 4.6 The validation of these predictions is described in Section 8.1.

Table 4.5: Preliminary propulsion package selected for all missions [1].

Motor	Battery Pack	Propeller
Sunnysky X4125 V3 420 K_v	6S2P LiPo Tattu 4500 mAh 6S, 25C	APC 15x10, E & EP

Table 4.6: Predicted performance from propulsion simulation (per motor).

Parameter	Prediction
RPM	8000
Thrust [Kgf]	5
Max Current [A]	75
Electrical Power [W]	1150

4.5 Initial Performance Estimates

Tables 4.7 and 4.8 provide an overview performance of the *Red Bearon* as predicted by KM Sim. Due to COVID restrictions, aerodynamic data gathered from the previous year was used for the KM Sim model. The data in this section is the same as the data in section 5.5 as there is no change from the initial to final performance estimates.

Table 4.7: Parameters for the *Red Bearon* KM Sim [1].

Flight Parameters	M1	M2	M3
CL Max (No Flaps)	1.2	1.2	1.2
CL Max (Flaps Deployed)	3.2	3.2	3.2
CD Cruise	0.05	0.05	0.06
L/D Cruise	9.4	7.5	12.6
Vcruise [m/s]	21	30	15
Vstall flaps deployed [m/s]	9.1	9.9	9.1
Vstall no flaps [m/s]	13.6	15.1	13.6
Takeoff Gross Weight [kg]	4.9	5.9	5.2

Table 4.8: The *Red Bearon* KM Sim performance.

Mission Parameters	M1	M2	M3	GM
Predicted Lap Time [s]	43.3	41.5	59.4	n/a
Number of Scoring Laps	3	3	10	n/a
Total Mission Time [s]	130	125	600	60
Number of Sensors	n/a	4	n/a	4
Score	1	1 + 0.63	2 + 0.4	0.5

5. Detail Design

5.1 Dimensional Parameters

Table 5.1 contains details of the wing and control surface geometries, while Table 5.2 provides data regarding the weight, take off distance, and other hindrances on the *Red Bearon* flight capabilities.

Table 5.1: Wing and control surface geometry [1].

Wing Geometry Data	
Wingspan [in]	60.0
Root Chord Length [in]	13.5
Tip Chord Length [in]	4.91
Wing Area [in ²]	561
AR	7.16
Leading Edge Sweep [°]	22.6
Quarter Chord Sweep [°]	20.4
H-stab Length [in]	15.9
V-stab Length [in]	7.78
Fowler Flap Length [in]	43.9
Elevator Area [in ²]	24.5
Rudder Area [in ²]	10.9
Aileron Area [in ²]	41.4

Table 5.2: Restraints and geometry of the aircraft [1].

Aircraft Restraints/Geometry	
Aircraft Empty Weight [kg]	5.64
Aircraft Payload Weight [kg]	0.90
Aircraft Maximum Weight [kg]	6.53
Cruise Velocity [m/s]	27
Available Static Thrust [kgf]	12.6
Available Dynamic Thrust at Cruise Velocity [kgf]	4.66
Take-off Distance [ft]	19.6

5.2 Structural Characteristics

The structure of the aircraft is composed of wood ribs connected by spars and stringers. The stringers are placed along the outer mold line of the airframe to increase the area moment of inertia of the wing and fuselage cross sections. To increase bending and torsional stiffness, carbon fiber (CF) spars are added to the wing and fuselage. Once completed, the structure is covered with UltraCote [1].

Spars are placed in the wing, fuselage, and empennage to ensure rigidity. Solidworks FEA was attempted for the placement of the spars. However, no computer was available that was capable of solving the complex geometry of the wing structure. As it is critical to keep the shear center as far forward as possible, the main spar in the wing is placed as far forward as possible while still extending through most of the wing span. To ensure both spars run uninterrupted through both sides of the wing, they are placed parallel to the Y direction rather than at the sweep angle. The secondary spar in the wing is placed further forward, allowing it to be thicker, but also not covering more than half of the span. Its placement allows the motor mounts to be placed at its ends. A single CF spar is also placed in the fuselage to ensure rigidity and provide a hard attachment point for the empennage and wing. In the empennage, wooden spars are placed at the thickest point in both [1]. The chosen combination is shown in Table 5.3.

Table 5.3: Spar placement and composition [1].

Part	Chord position (Center)	Material	Outer Diameter [in]	Inner Diameter [in]	Thickness [in] (square)
Wing	0.29	Carbon Fiber	0.715	0.625	n/a
Wing	0.58	Carbon Fiber	0.49	0.375	n/a
Fuselage	n/a	Carbon Fiber	0.34	0.25	n/a
HStab	0.4	Basswood	0.25	n/a	n/a
VStab	n/a	Balsa	n/a	n/a	0.25

5.3 System and Subsystem Design

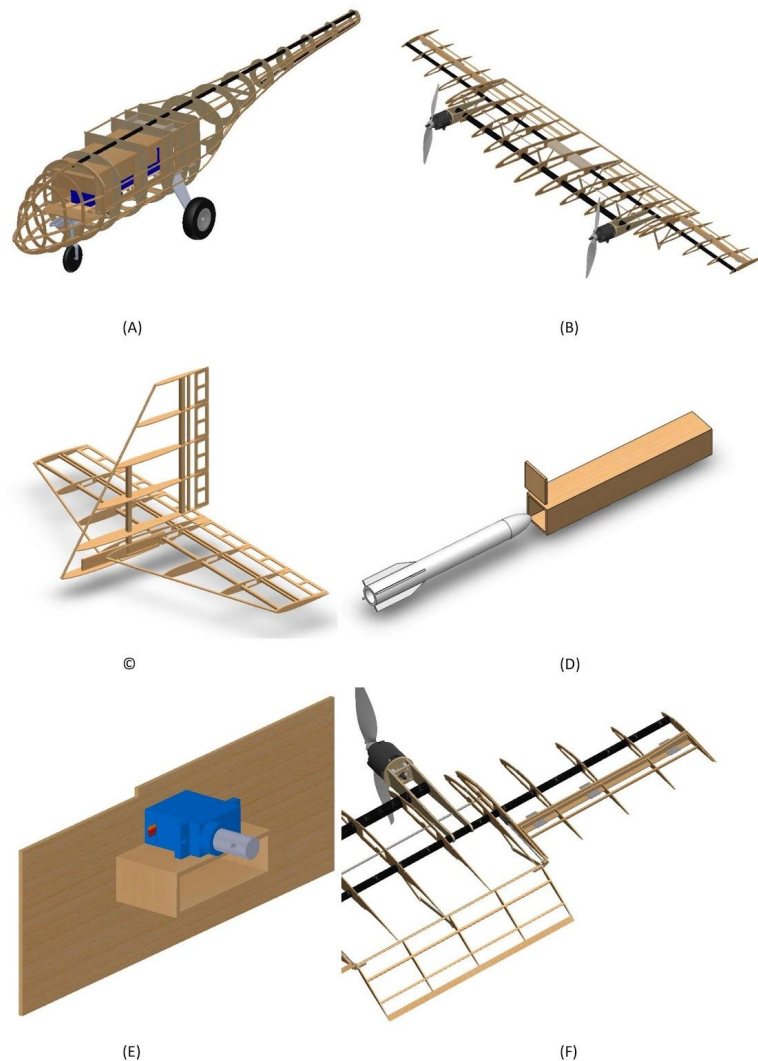


Figure 5.1: CAD renderings of (A): Fuselage, (B): Wing, (C): Empennage, (D): Sensor and container, (E): Sensor mechanism, (F): Deployed Fowler flap.

5.3.1 Fuselage

The fuselage stores electronic components, carries payloads, and connects the wing and tail. Bulkheads are made from Lite plywood. The nose cone is split in top and bottom halves. The upper half is

connected to the fuselage by a hinge to allow access to the avionics and passenger compartments, while the lower half is rigidly connected to the front bulkhead. The largest section of the fuselage is a compartment designed to hold the batteries and three sensor containers. A CF spar runs lengthwise through the fuselage, increasing stiffness and providing a mounting point for the tail. To save weight and create wire pathways, excess material is cut from the bulkheads and attachment plates. The fuselage is shown in Fig. 5.1A.

5.3.2 Wing

The wing consists of 21 balsa ribs with two CF spars. The spars are positioned as far forward as possible to move the shear center in front of the aerodynamic center. To increase main wing leading edge stiffness and ensure that the curvature was correct between the ribs, the front quarter chord of the wing is wrapped with a $\frac{1}{16}$ in. balsa skin. The leading edge is a square balsa spar inset into the ribs and sanded down to match the curvature of the airfoil. Because of the sweep of the wing and decreasing thickness of the ribs, it is impossible to run the CF spars through the entire wing. The largest spar has a diameter of 0.715 in. and runs through seven ribs, while the smaller spar has a diameter of 0.49 in. and runs through seventeen ribs. To stiffen the outer section of the wing, a $\frac{1}{8}$ in. balsa beam runs through the outer five ribs on either side. This is oriented vertically to increase the area moment of inertia, and is a mount for the hinges of the aileron. Both motors are mounted directly next to the ends of the thickest spar. This gives a minimum distance from the center of the motor to the fuselage of 8.6 in., allowing for a maximum 16 in. diameter propeller. The motor mounts are made of Lite plywood, and the motor is placed in front of the leading edge of the wing while linking to both of the CF spars, distributing the load. The wing mount consists of two plates, one in the wing and one on the top of the fuselage to allow for removal. The plates are attached using two 10-32 bolts, which screw into T-nuts in the plate on the fuselage. The bolts are placed 4 in. apart in the Y-axis to maximize the reaction moment. A plywood beam extends past the leading edge of the wing in the middle, and slots into a bracket connected to the spar in the fuselage, providing three points of contact for the wing [1]. The wing is shown in Fig. 5.1B.

5.3.3 Empennage

The empennage is designed using the same rib construction as the main wing. Due to the smaller scale of both the vertical stabilizer and horizontal stabilizer, a $\frac{1}{4}$ in. diameter birch spar is added to the horizontal stabilizer for rigidity. A $\frac{1}{4}$ in. square balsa spar is placed in the vertical stabilizer for rigidity and mounting the rudder hinges. The birch spar is used to connect the horizontal stabilizer to a plate at the base of the vertical stabilizer, and the bottom rib of the vertical stabilizer is connected to the main spar of the fuselage. The attachment points to the bulkheads are each spaced so that the horizontal stabilizer is attached at a 1° incidence angle [1]. The empennage is shown in Fig. 5.1C

5.3.4 Landing Gear

The initial design of the aircraft included retractable landing gear to reduce drag in flight. To avoid the challenges of manufacturing retractable landing gear, the actuators and struts were purchased from Robart Manufacturing. The actuators were rated for 12-18 lb aircraft, which is sufficient for the AUW. The nose gear strut is steerable using a servo. The main gear were originally attached to CF square rods in the fuselage using birch plywood brackets. The nose gear actuator was connected to a birch plywood plate attached to the front bulkhead. To increase the stability of the gear while rolling, the nose gear was angled forward and the main landing gear was towed in [1].

Through testing of the aircraft the previous year in 2020, several inadequacies of the initial landing gear design were discovered (see section 8.5). The wooden brackets mounting the actuators to the CF rod proved very prone to failure, even when manufactured with birch plywood. Also, upon landing during the first flight test, the CF rod itself cracked along the axial direction, likely due to being a pultruded rod and therefore weaker in torsion. Consequently, it was decided that, due to time constraints and difficulties manufacturing the correct components, the landing gear would no longer be retractable. To reduce the weight and resolve the mounting issue, the next iteration of the main landing gear will be a solid-spring type while the nose gear remains unchanged. Two materials were considered for the strut, aluminum and carbon fiber. The strut was designed using sizing equations for solid spring landing gear, Euler-Bernoulli Beam theory, and estimating a maximum landing deceleration of 3g as the first test flight landed at 1g max [14]. The aluminum weighed 100 g with a factor of safety of 3. Using Solidworks Finite element analysis (FEA), the maximum von Mises stress was 258 MPa. With a yield stress of 505 MPa (7075T6 aluminum), the plate would have a final safety factor of approximately 2 [1]. These decisions made for the 2020 aircraft carried over to the 2021 aircraft used in this report.

Despite the favorable results with the aluminum, an equivalent CF strut met the same sizing requirements while weighing less. Carbon fiber was selected over aluminum due to its lower weight, wider wheel span, and significantly lower manufacturing time [1].

5.3.5 Sensor Storage

The sensors are stored above the battery within shipping containers, all of which is contained within the main body of the fuselage. Each container is constructed with $\frac{1}{16}$ in. lite ply shown in Fig. 5.1D. The container acts as a box with one of its end faces open for the insertion of a sensor. Once the sensor is inserted, the container is closed with a square lid. For the configuration for the payload mission, three sensor containers are stored above the battery with minimal spacing, but are secured by the constraints of the inner walls of the fuselage.

5.3.6 Sensor Release Mechanism

The sensor release mechanism is positioned below the fuselage, behind the nose landing gear. The mechanism (Fig. 5.1E) consists of two components: the mount and the reeling mechanism. The mount includes a thin panel attached to the side, top, and bottom of the casing of its compartment.

Attached to this panel is an elevated support that holds a servo responsible for driving the reeling mechanism. The reeling mechanism consists of a customized cylindrical axle fit to the servo that is modified to continuously rotate. The axle contains a small hole for attaching the tow cable. The rotation of this axle will control the recovery and deployment of the sensor by winding and unwinding the tow cable around the axle.

5.3.7 Flap Mechanism

This design uses Fowler flaps, which must achieve the correct geometry to maximize lift while rigid enough to handle the maximum motor thrust. The flaps are made from balsa ribs with basswood spars and balsa plates to increase the rigidity enough to be covered with UltraCoat without warping. The flap is attached to a slide on either side. These are attached to the ribs on either side of the flap. This allows the correct amount of translational motion in the Fowler flap. To achieve the rotational motion three hinges are attached, one to either side and one in the center. The hinges are made from two Lite plywood beams of varying lengths. The middle hinge has a slightly shorter beam, causing it to “pull” on the flap as it extends, rotating it to 39° deflection. The hinges are rigidly attached to the same $\frac{1}{8}$ in. steel rod rotated by a servo. The lengths of the beams required to achieve the correct geometry were verified using a SolidWorks motion study [1]. The flap mechanism is shown in Fig. 5.1F.

5.3.8 Electrical Diagram

Figure 5.2, an electrical schematic of the whole aircraft, clarifies the arrangement and interaction of the various electrical subsystems.

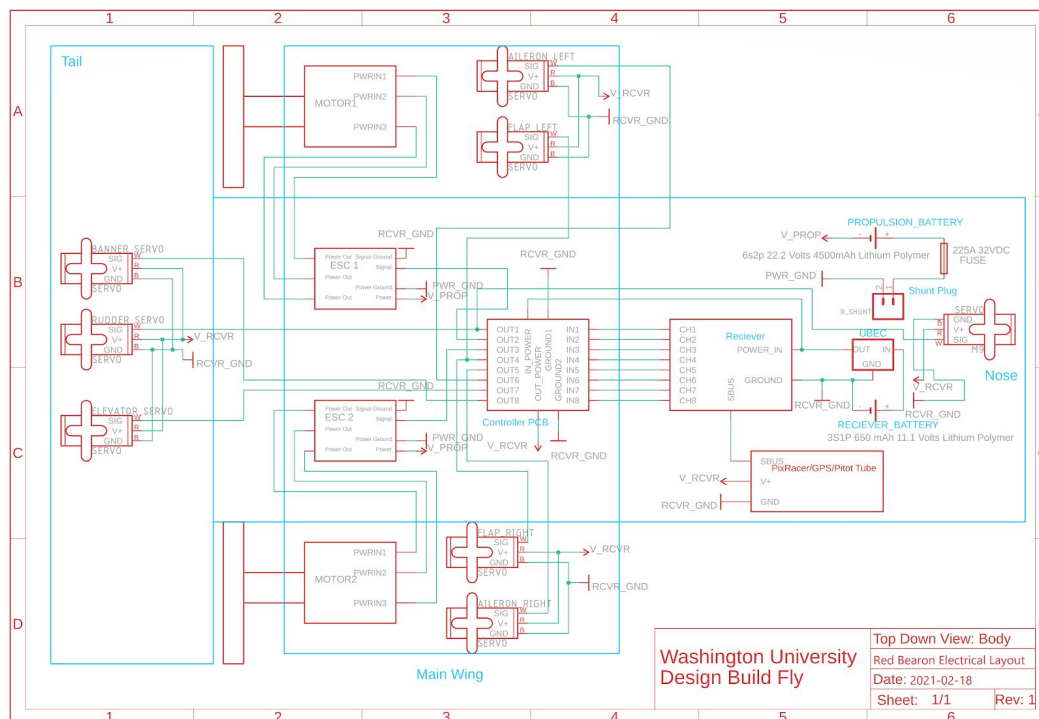


Figure 5.2: Electrical layout of the Red Bearon [1].

5.3.9 Servo Selection

Servos were selected based on torque requirements determined from maximum aerodynamic hinge moment predictions in XFLR5 using a fast, inviscid method. A MATLAB script was then written to predict moments on a servo horn given the control surface hinge moment and the system geometry. An additional factor-of-safety was applied on top of the calculated torque to account for uncertainties. Servo meeting the torque requirements were then selected based on weight and response time [1]. The final chosen servos for each mechanism are outlined in Table 5.4.

Table 5.4: Chosen servos, torques, and quantity on each mechanism [1].

Part	Servo	Quantity
Ailerons	BMS-A53M HV	2
Rudder	DS-929HV	1
Elevator	BMS-101HV	1
Flaps	DS-929HV	2
Nosewheel	DS-929HV	1
Sensor	DS-929HV	1

5.3.10 Propulsion Package

As explained in Section 4.4, two Tattu 4500 mAh 6S, 25C LiPo batteries connected in parallel were chosen to provide the necessary energy to last the full 10 minute M3. A two motor configuration was selected based on superior efficiency, a reduced need for ground clearance, and ease of container loading. From the propulsion simulations outlined in Section 4.4, two Sunnysky X4125 V3 420 K_v motors each with an APC 15x10E propeller were selected to provide adequate power and maximize flight speed at cruise. A T-motor 80 Amp T80A ESC was selected to handle the maximum predicted current draw [1]. The final selected propulsion package is shown in the table below:

Table 5.5: Chosen propulsion components [1].

Component	Selected
Motor	Sunnysky X4125 V3 420 K _v
Battery Pack	2 x LiPo Tattu 4500 mAh 6S, 25C
Propeller	APC 15x10, E & EP
ESC	T-motor 80 Amp T80A ESC

5.4 Mass and Balance

Each mission has a different mass distribution. To estimate the CG of the vehicle, component masses and locations were determined from the CAD model and tabulated. Once found, the components may be referenced using absolute coordinates in the traditional body axis. The tip of the nose, was taken to be the origin, and a mutually orthogonal coordinate system was established, with the positive X-axis pointing out from the nose of the aircraft, positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right wing, relative to the nose facing forward. The masses and absolute

coordinates of the aircraft's major components are listed in Table 5.6. Note that components such as bolts, glue, covering, and wire mass are not included in the table [1].

Table 5.6: Mass and CG table [1].

Mission	Mass [kg]	X [in]	Z [in]
Wing Structure	.41	-21.8	-5.68
Fuselage Structure	.19	-22.2	-2.19
VStab Structure	.021	-56.1	-6.11
HStab Structure	.015	-57.2	-4.75
Landing Gear (Main)	0.27	-21.6	0.67
Landing Gear (Nose)	0.19	-8.46	1.46
Electronic Components	0.64	-15.8	-1.4
Main Battery	1.64	-16.7	-0.94
Motors	0.83	-14.9	-5.18
M1 (empty)	4.24	-17.2	-2.17
Sensor Mechanism (Empty)	.036	-12.7	0.63
M2	4.44	-17.1	-2.19
Sensor Containers (Full)	.195	-14.6	-2.58
Sensor Mechanism (Empty)	.036	-12.7	0.63
M3	4.30	-17.3	-2.13
Sensor Mechanism (Full)	.091	-18.5	0.63

5.5 Flight and Mission Performance

Table 5.7 provides aerodynamic properties and Table 5.8 provides performance data synthesized from flight tests and KM Sim updated with experimental wind tunnel and thrust data. Data was normalized using other aircraft created in section 3.4. Ground Mission score was estimated based on best guess layouts and times for other aircraft and other teams' times [1].

Table 5.7: Parameters for the Red Bearon KM Sim [1].

Flight Parameters	M1	M2	M3
CL Max (No Flaps)	1.2	1.2	1.2
CL Max (Flaps Deployed)	3.2	3.2	3.2
CD Cruise	0.05	0.05	0.06
L/D Cruise	9.4	7.5	12.6
Vcruise [m/s]	21	30	15
Vstall flaps deployed [m/s]	9.1	9.9	9.1
Vstall no flaps [m/s]	13.6	15.1	13.6
Takeoff Gross Weight [kg]	4.9	5.9	5.2

Table 5.8: The *Red Bearon* KM Sim performance.

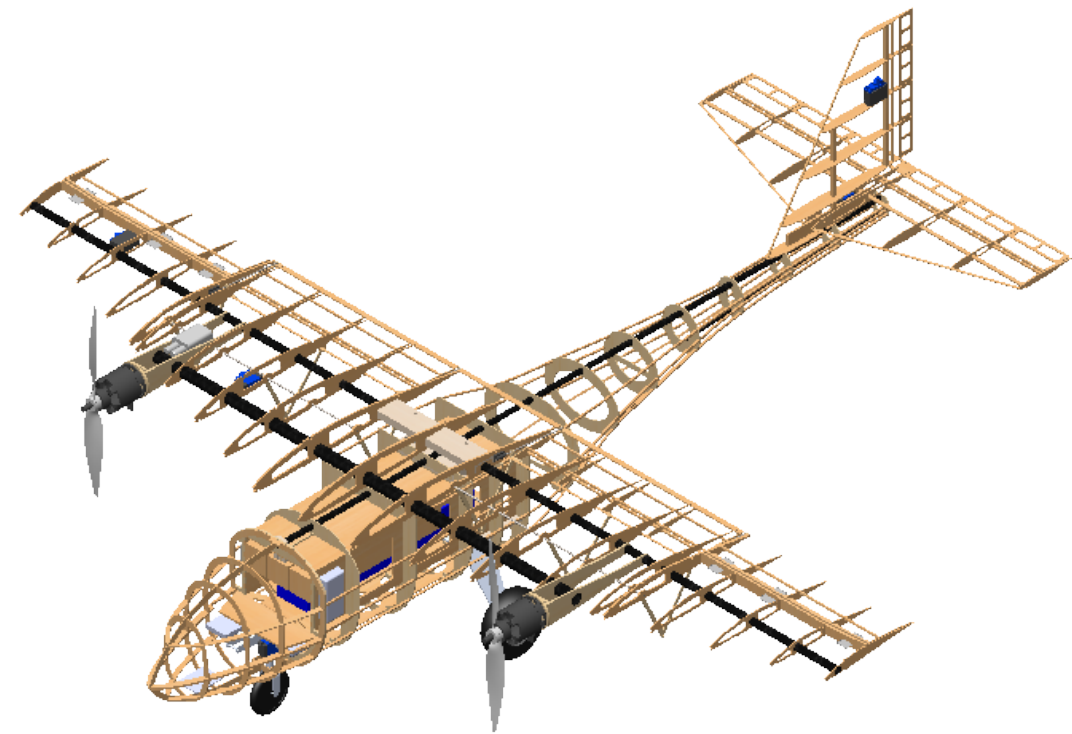
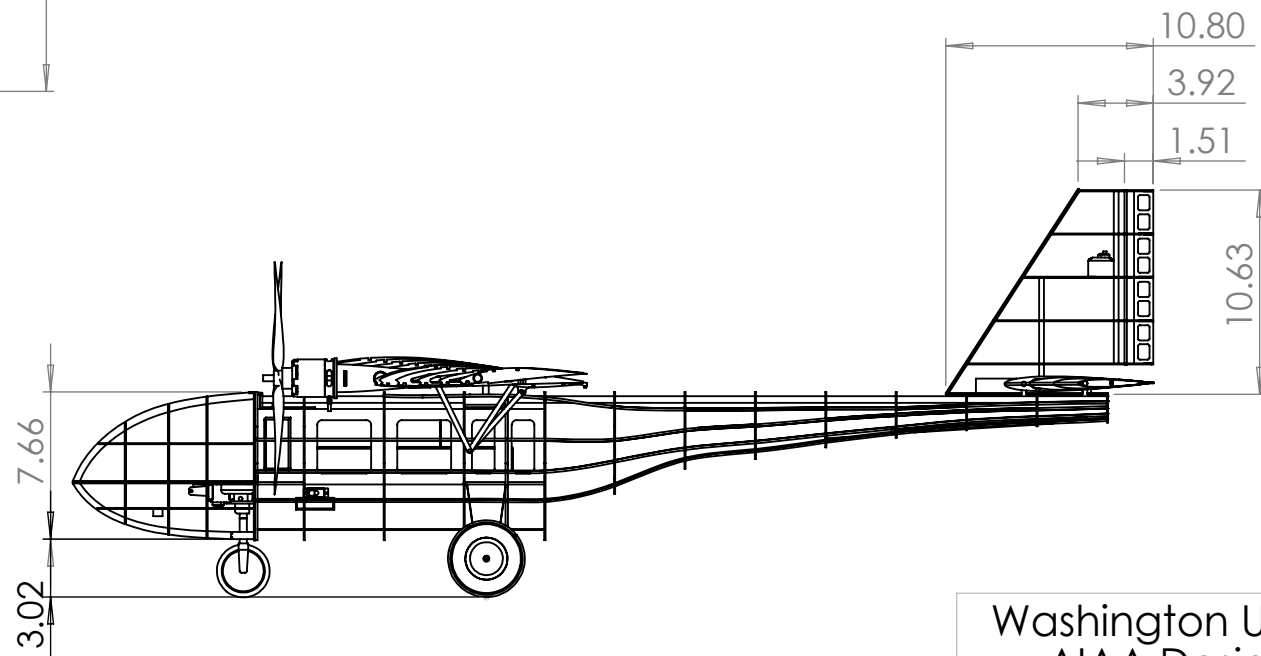
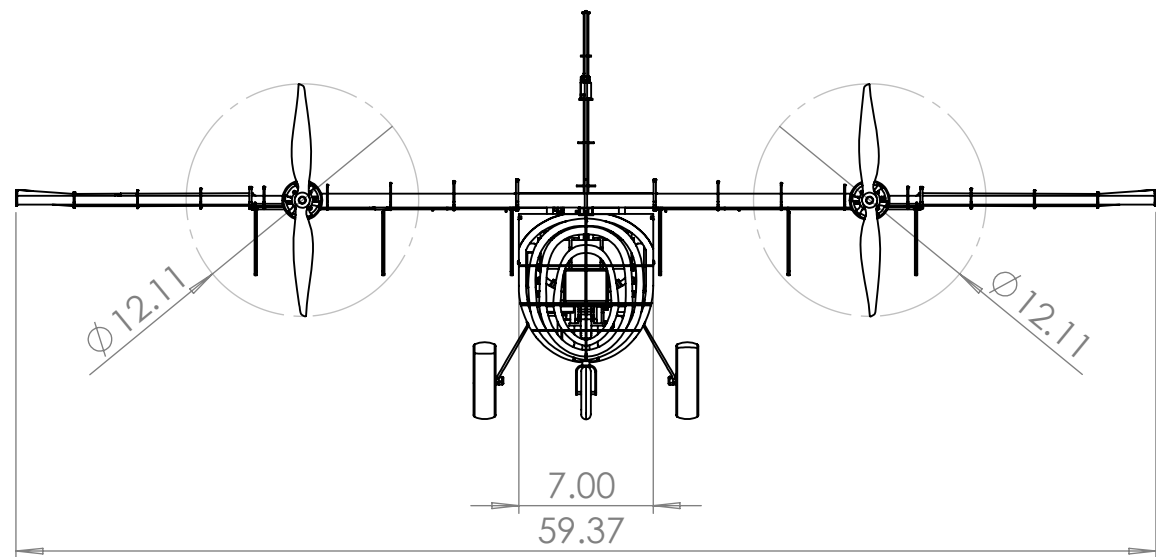
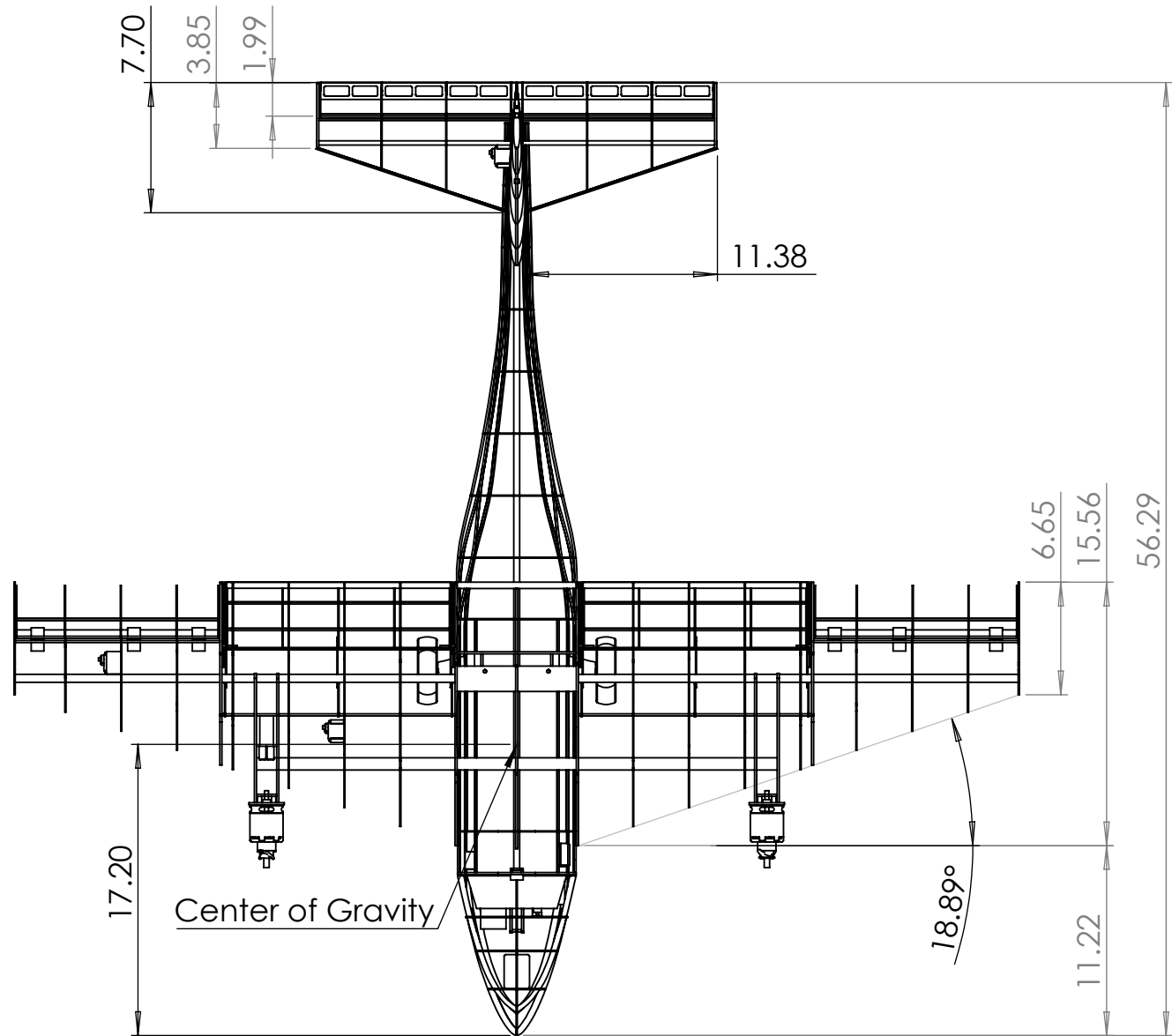
Mission Parameters	M1	M2	M3	GM
Predicted Lap Time [s]	43.3	41.5	59.4	n/a
Number of Scoring Laps	3	3	10	n/a
Total Mission Time [s]	130	125	600	60
Number of Sensors	n/a	4	n/a	4
Score	1	1 + 0.63	2 + 0.4	0.5

5.6 Drawing Package

Detailed in the following pages is the drawing for the *Red Bearon*. Drawings are, in order, three-view, structural layout, systems overview, and layout of payload accommodations.

B

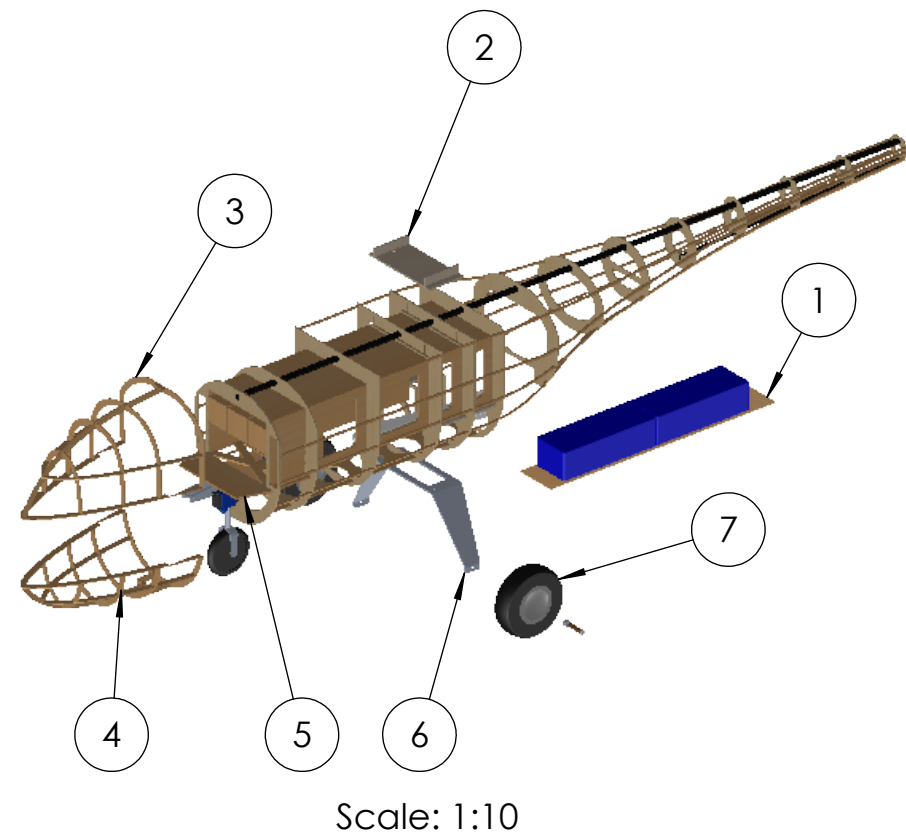
A



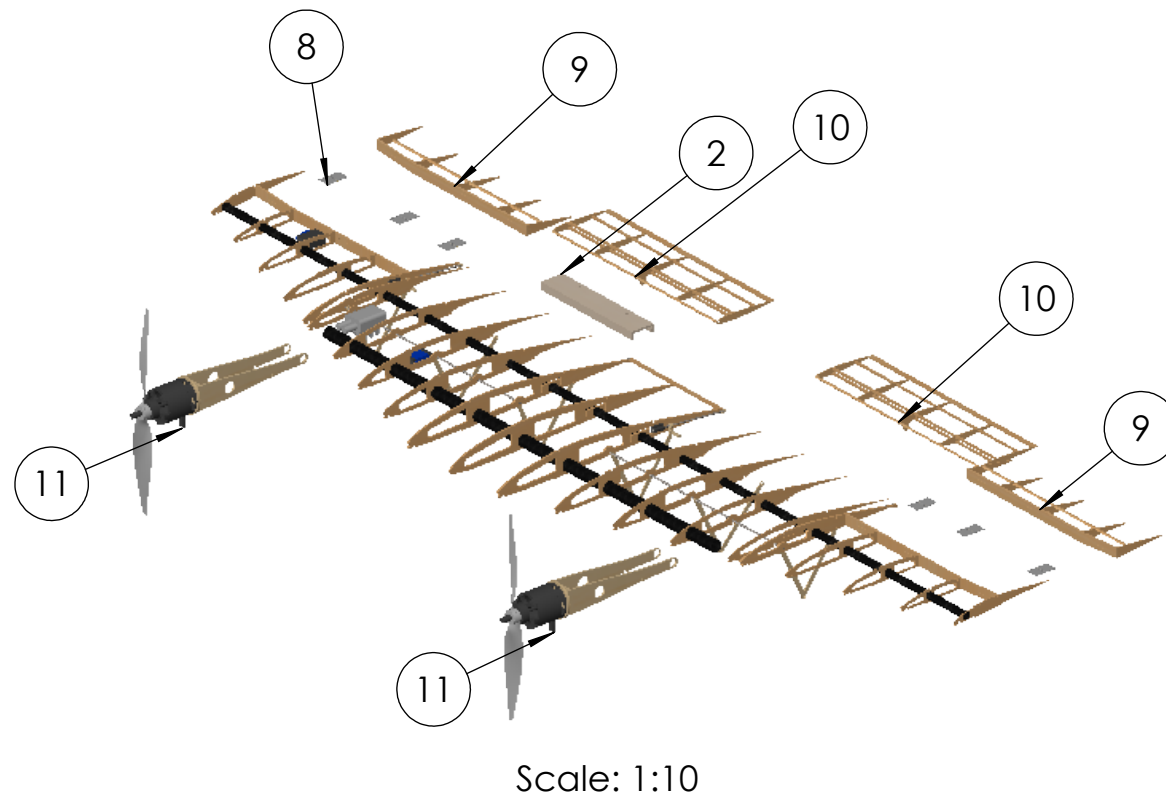
B

A

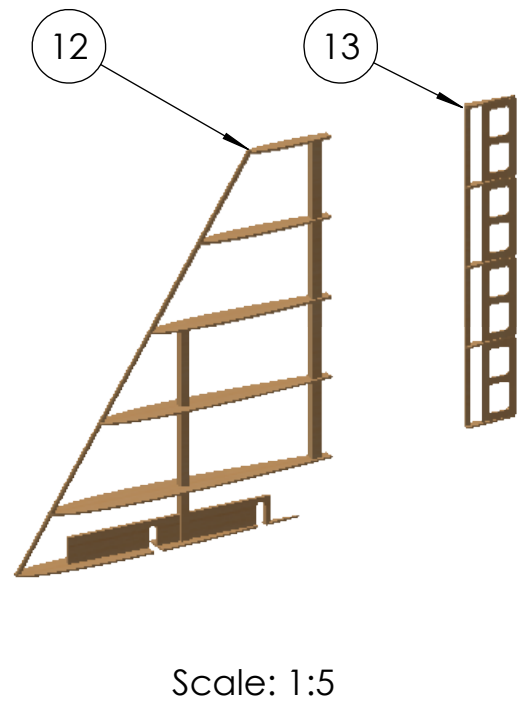
B



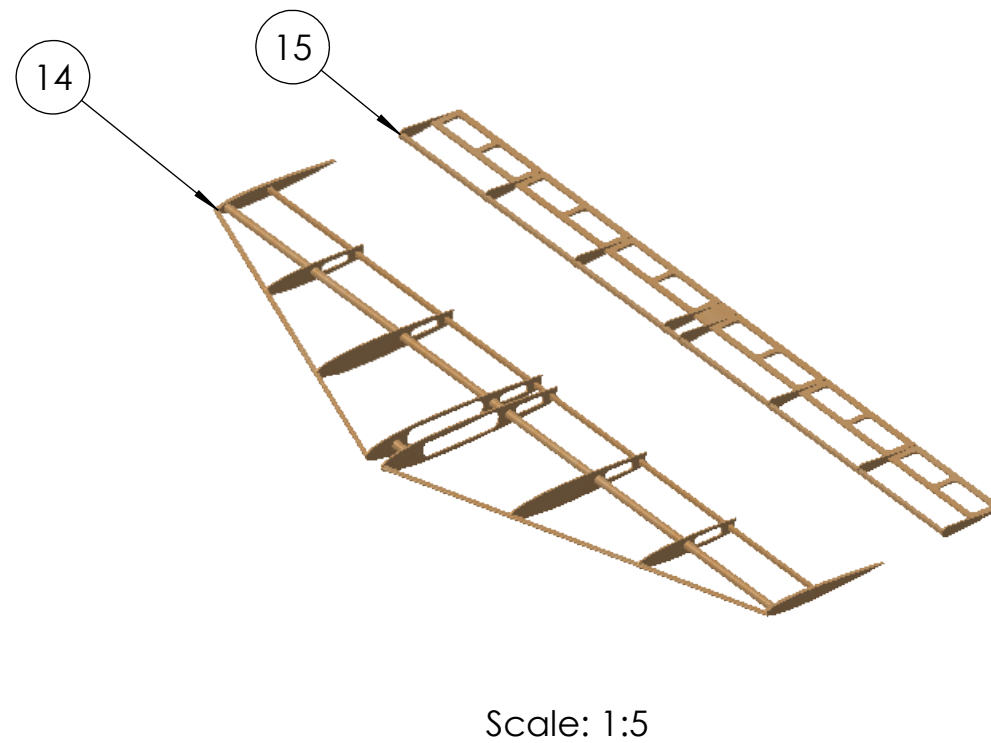
B



A



A

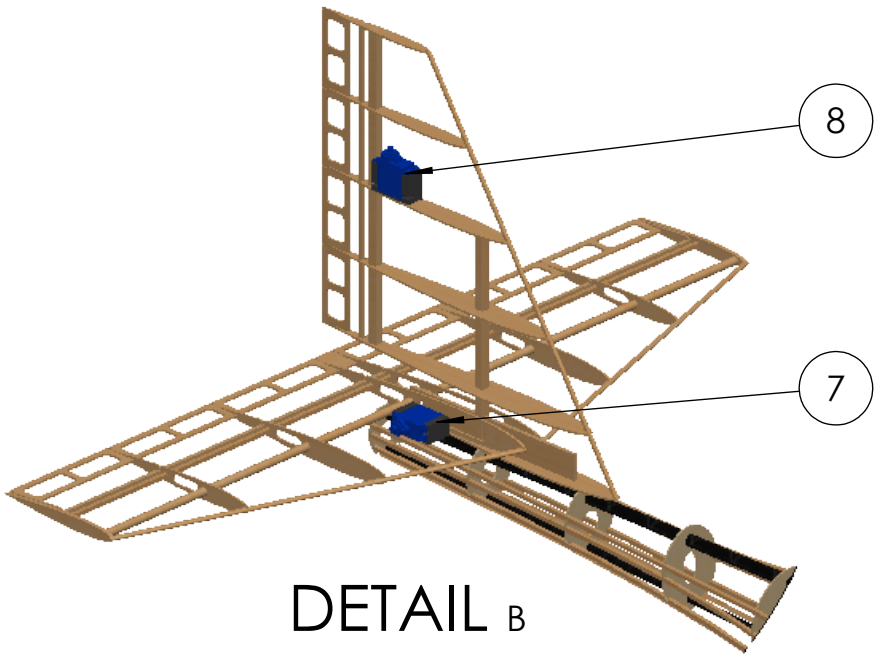


Item	Item Name
1	Battery Mount
2	Wing-Body Connection
3	Upper Nose Cone
4	Lower Nose Cone
5	Avionics Mount
6	Landing Gear Strut
7	Main Landing Gear Wheel
8	Hinge
9	Aileron
10	Fowler Flap
11	Motor Mount
12	Vertical Stabilizer
13	Rudder
14	Horizontal Stabilizer
15	Elevator
Washington University in St. Louis AIAA Design/Build/Fly 2021	
TITLE: Red Bearon	DWG. NO. Structural Layout
SHEET 1 OF 1	

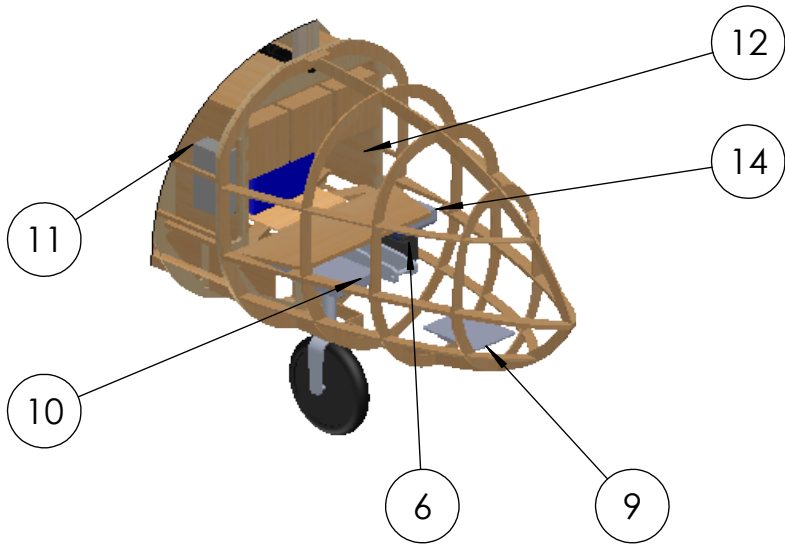
B

A

B



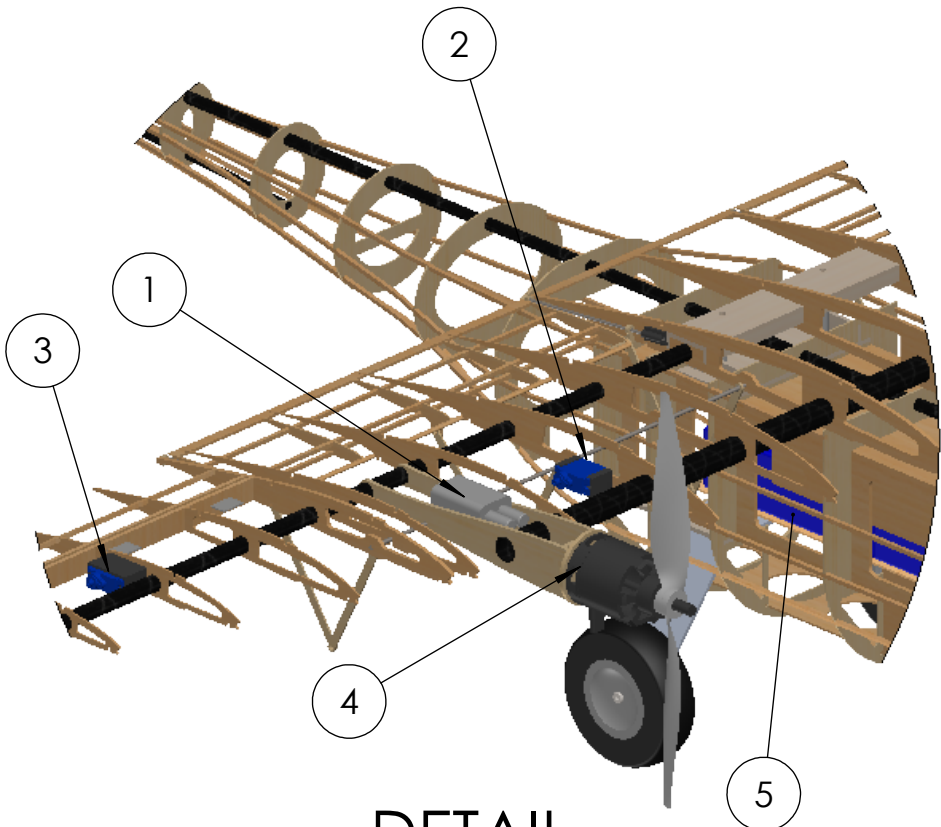
DETAIL B
SCALE 1 : 5



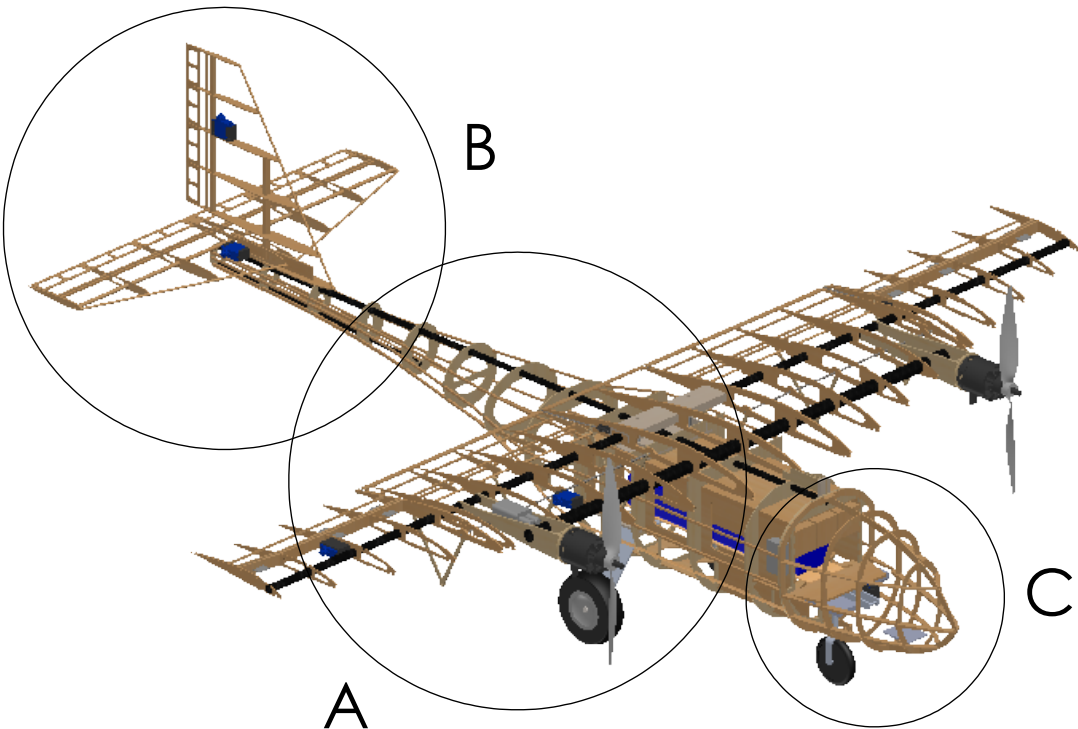
DETAIL C
SCALE 1 : 5

B

A



DETAIL A
SCALE 1 : 5



A

B

C

A

Item	Part Name	Manufacturer
1	80A ESC	T-motor
2	Fowler Flap Servo	Corona
3	Aileron Servo	Blue Bird
4	420Kv Motor	Sunny Sky
5	6s 9000mAh BAttery	Genstattu
6	Nose Wheel Servo	Corona
7	Elevator Servo	Corona
8	Rudder Servo	Blue Bird
9	GPS	MRobotics
10	Servo Control Board	Arduino
11	Reciever	FrSky
12	3s 650mAh Battery	Genstattu
13	Pitot Tube	Pixhawk
14	UBEC	Hobbywing

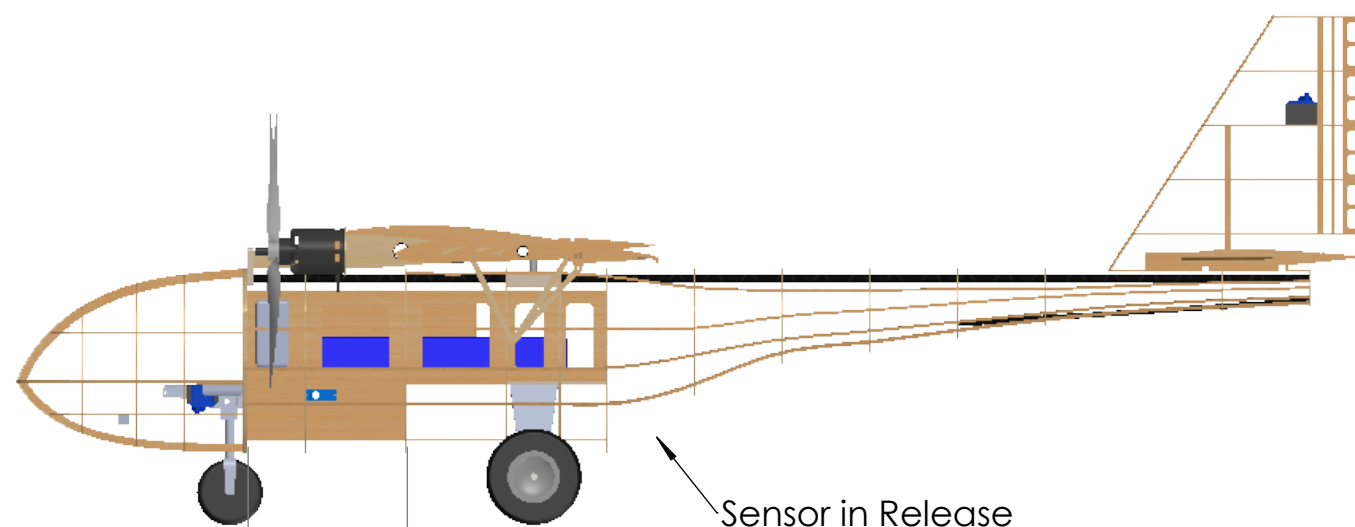
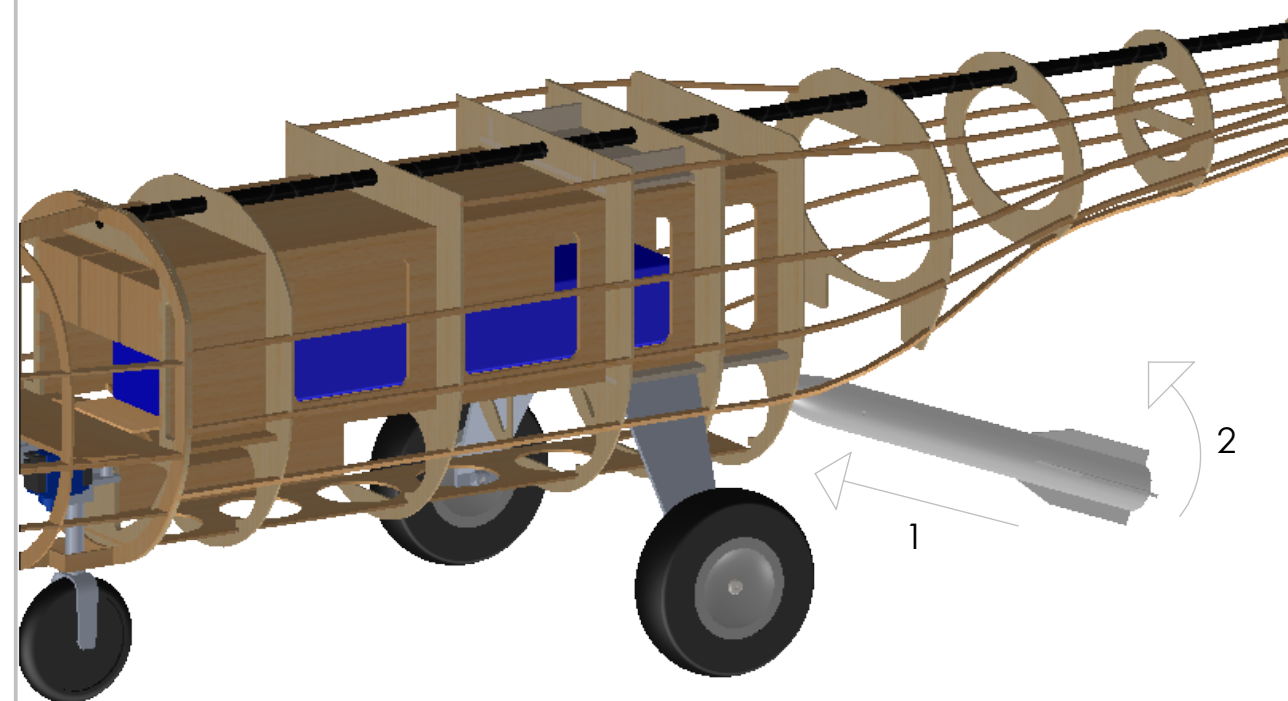
Washington University in St. Louis AIAA Design/Build/Fly 2021		
TITLE: Red Bearon	DWG. NO. Systems Overview	
SCALE: 1:20		SHEET 1 OF 1

4

3

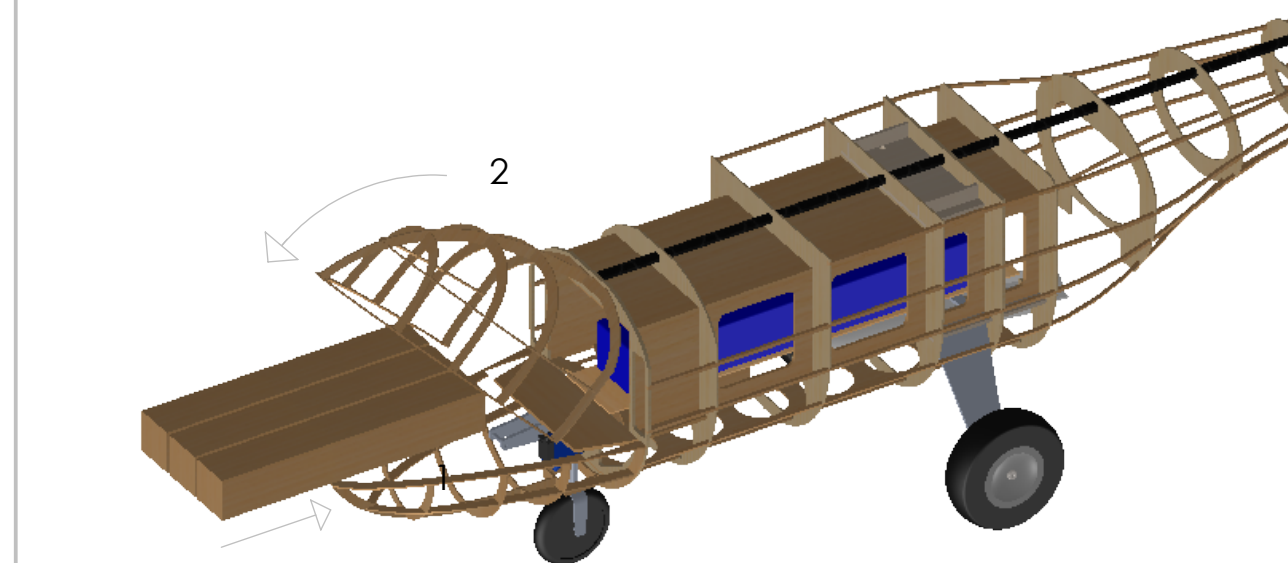
2

1

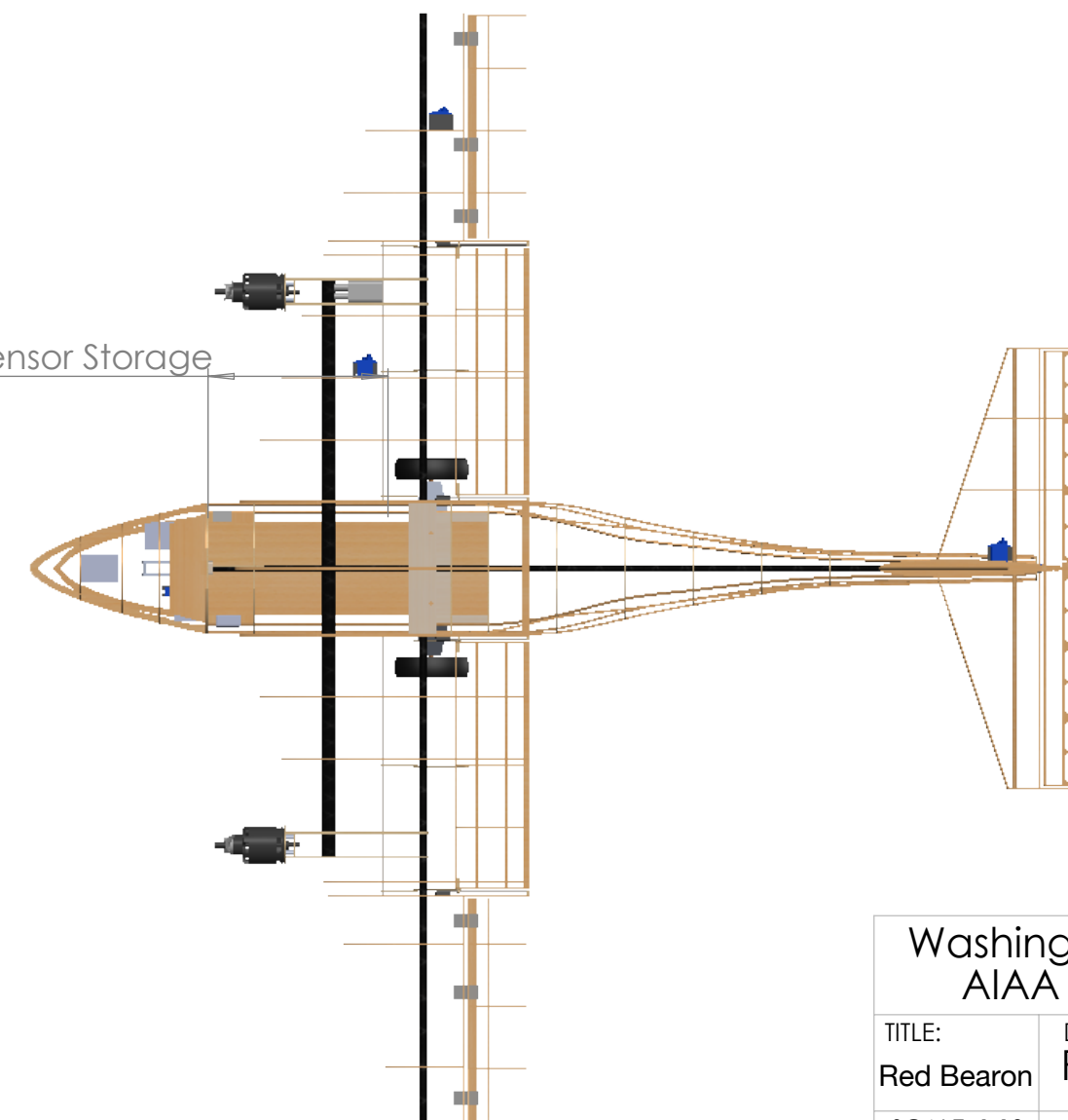


Sensor Release Mechanism

Sensor in Release



Sensor Storage



Washington University in St. Louis
AIAA Design/Build/Fly 2021

TITLE:
Red Bearon

DWG. NO.
Payload Accomadation

SCALE: 1:10

SHEET 1 OF 1

3

2

1

6. Manufacturing Plan

6.1 Manufacturing Methods Considered on Major Components

6.1.1 Selection Criteria

Decreasing manufacturing risk played a significant role in the development of this year's aircraft. As such, the Manufacturing Division sought a manufacturing method which enables rapid prototyping while still flexible and forgiving. Thus the importance of Ease of Iteration was set to 0.25. Cost of prototyping was also taken into consideration, with a weight of 0.10. For issues not immediately caught, flexible construction methods would allow for adjustments, maintenance, and repairs. Flexibility was thus determined to be 0.20. These goals were considered along with expected strength, weight, reliability, and team experience, with importances 0.15, 0.10, 0.15, and 0.05, respectively [1].

6.1.2 Solid Foam

The Ease of Iteration was scored at 2. Damaged foam construction cannot be repaired and must be replaced, meaning a Flexibility score of 2. Finally, a fuselage with a small cross section cannot be structurally supported by foam: Expected Strength and Reliability were scored at 4. Expected Weight and Cost scores were estimated as 5 and 4, respectively. As WUDBF built a foam plane two years ago, it received an Experience score of 8 [1].

6.1.3 Additive Manufacturing

The availability of 3D printers makes it appealing. Its rapid prototyping ability warranted an iteration score of 7 and a Flexibility score of 4. However, 3D-printed structural components have historically failed. The printing process was also unreliable and led to delays. Components were heavy, warp-prone, and split between print layers. It therefore received strength, reliability, and weight scores of 0. Past experience led to an Experience score of 8. Expected Cost was scored at 5. Additive manufacturing was thus only considered for small, nonstructural components [1].

6.1.4 Monocoque Methods

Fiberglass or CF monocoque methods were considered but ultimately dismissed due to their high costs, difficulty of maintenance, lack of team experience, and an inability to iterate quickly. The method scored low in these categories, as shown in Table 6.1. However, prefabricated CF rods and fiberglass as reinforcement were pursued further. Tests show that they could strengthen the wing and fuselage. As such, strength, weight, and reliability scored at 8, 5, and 5, respectively. The wing proved to be strong enough without fiberglass, so it was not needed [1].

6.1.5 Wood Construction

Wood construction was previously researched and used in last year's aircraft. It proved to be relatively cheap, light, easy to work with, and readily available. Alterations and repairs could be made easily. Access to a laser cutter enabled rapid prototyping. Furthermore, we had a largely complete

airframe built last year that we could adapt for this year's aircraft. These factors resulted in high scores in all categories except experience. Exact scores in each category are shown in Table 6.1 [1].

6.1.6 Methods Chosen for Major Components

A summary of the manufacturing methods is shown in Table 6.1.

Table 6.1: Score table for manufacturing methods [1].

Feature	Importance	Method			
		Solid Foam	Additive Manufacturing	Monocoque	Wood
Ease of Iteration	0.25	2	7	3	8
Flexibility	0.20	2	4	2	8
Expected Strength	0.15	4	0	8	7
Expected Reliability	0.15	4	0	5	8
Expected Weight	0.10	5	0	5	5
Expected Cost	0.10	4	5	2	7
Experience	0.05	8	8	0	8
Weighted Total Score	1	3.4	3.45	4.2	11.25

Wood construction was chosen as the primary manufacturing method for all major components (Table 6.1) due to its high score. The shape of the wing and fuselage were defined by laser cut ribs. Wood stringers ran the length of each. Both feature CF rods to increase rigidity. The tail assembly and control surfaces were made with balsa ribs and stringers. Landing gear was purchased from a third party and attached with birch ply mounts [1].

6.2 Manufacturing Process

Due to COVID-19 restrictions, construction of the 2020-2021 aircraft is not complete. Most photos in this section are of the 2019-2020 aircraft as it shows the complete manufacturing process of the airframe, photos of the current status of the *Red Bearon* airframe are shown at the end of the wing, fuselage, and tail sections.

6.2.1 Wing

The basic elements of the wing are the ribs and stringers. Ribs must be positioned correctly before stringers are installed: U-shaped jigs secure them during construction. The central 13 ribs were installed to allow the load bearing CF rods to be inserted. The motor mount pieces attach to the CF rods and so were also inserted. The remaining ribs were then similarly installed. Basswood stringers were added for structure and balsa stringers for the leading edge, where it was sanded to shape. Thin wing sections above the Fowler flaps were reinforced with balsa sheets, shown at right in Fig 6.1 [1].

After the frame was completed, servos and control surfaces were installed, the jigs were cut off and sanded down, and the wing was flipped over. Remaining structural basswood sticks were added and the wing attachment plate was inserted. The motors were integrated at this stage. The leading edge was formed by wrapping balsa sheets soaked in warm water around a stringer, as determined in section 8.3.1.

Sharper portions of the leading edge used two smaller sheets of balsa. A smoother UltraCote surface was achieved with additional balsa sticks [1]. The complete wing is shown at left in Fig. 6.1.

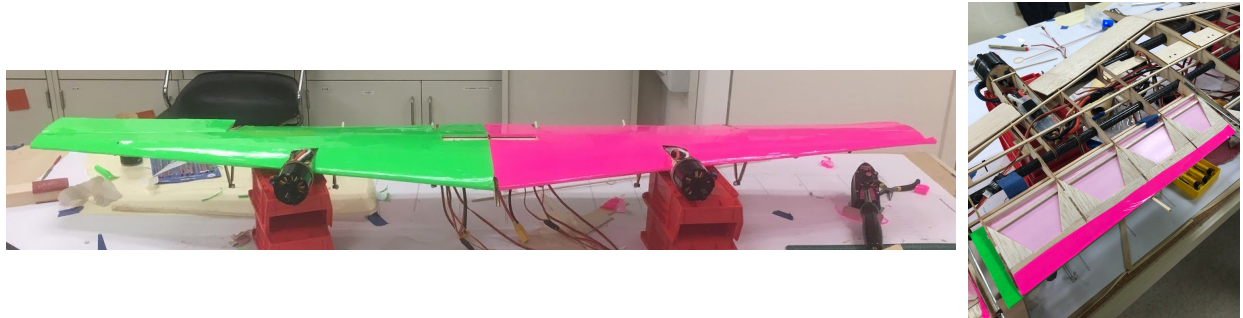


Figure 6.1: Left: complete main wing. Right: Fowler flap area trailing edge reinforcements [1].

6.2.2 Control Surfaces and Attachments

The *Atlas* has two ailerons, two Fowler flaps, a rudder, and an elevator. The main body of each was constructed using jigs for proper alignment. The frames for each control surface feature lateral basswood stringers. Trailing edges feature 1/16 in. balsa sheets sanded to half width. The ailerons also feature a sanded leading edge which matches the airfoil. Holes were cut into the leading edges (except on the Fowler flaps) to house plastic hinges which enable free movement of the control surface. Strength testing showed additional structure was required, accomplished with balsa sheeting and cross bracing [1].

Figure 6.2 shows the completed Fowler flap. Each Fowler flap had three arms linking the flap to a steel dowel. The outside arms are connected to a pin which slides along a rod as the Fowler flap extends. The arms segments were connected by a screw and nut. The team added wooden attachments to the arms to keep the Fowler flaps secure while stowed [1]. UltraCote was applied before the control surfaces were installed, shown at right in Fig. 6.2.

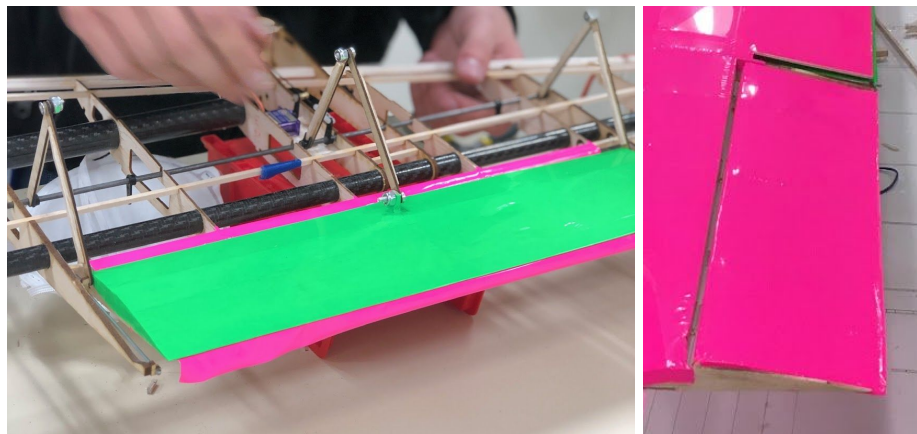


Figure 6.2: Left: coated Fowler flap. Right: coated aileron [1].



Figure 6.3: Current status of 2020-2021 wing.

6.2.3 Fuselage

The fuselage is constructed from birch ply ribs and basswood stringers. Birch ply was chosen for its strength, and basswood was chosen for its higher relative strength and adequate flexibility—the fuselage design features curved stringers. A jig ensured proper rib alignment. Stringers were added from the tail forward to connect ribs. Some stringers were spliced for additional length. Once in place, the team integrated the CF rods described in section 5.3.1. Next, platforms for the electronics and passengers were installed, shown in Fig 6.4. Finally, the fuselage was streamlined with UltraCote; however, this was done last in the manufacturing process to simplify systems integration [1].

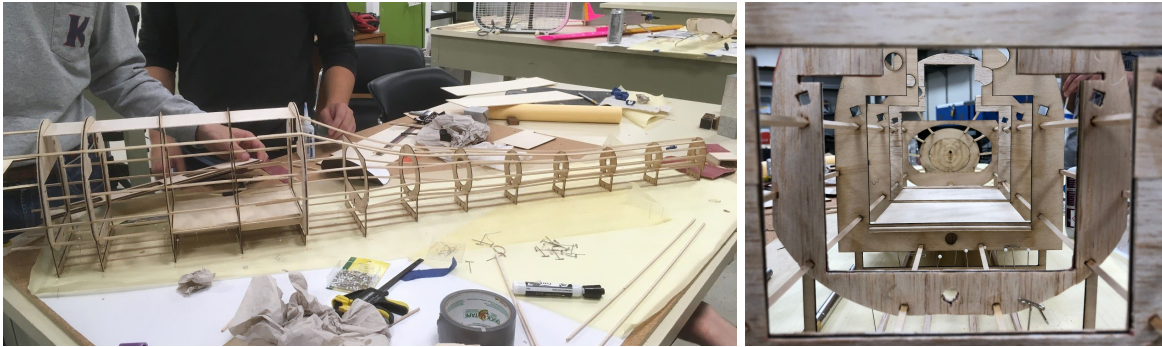


Figure 6.4: Left: fuselage construction. Right: bare fuselage interior [1].

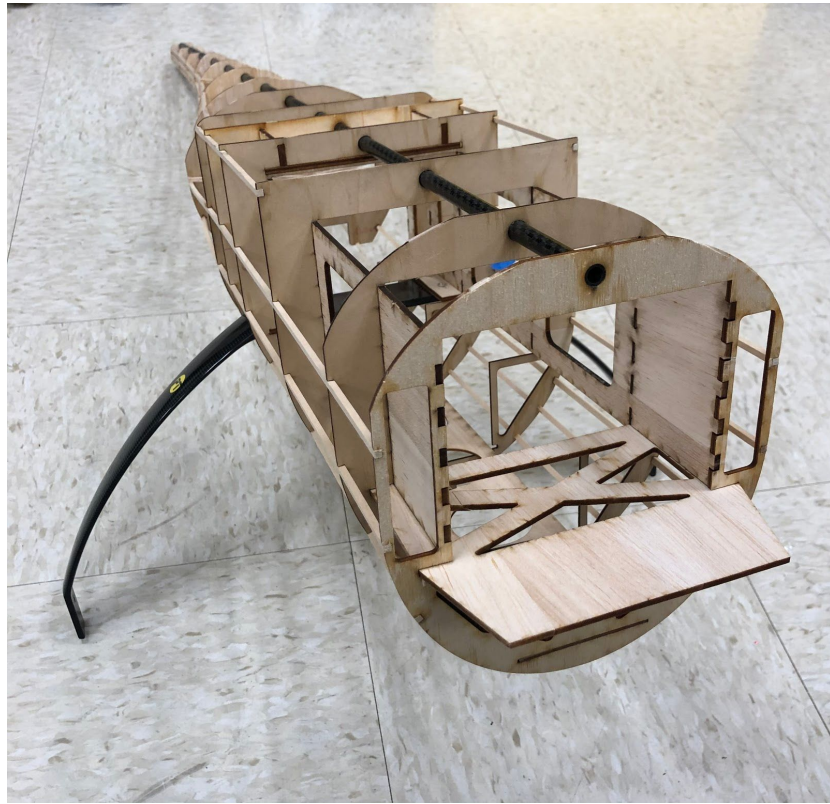


Figure 6.5: Current status of 2020-2021 fuselage.

6.2.4 Tail

The tail was constructed using basswood stringers, balsa wood, wooden dowels, and birch ply. Birch ply was used as the connection point to the fuselage. Basswood stringers served as the leading edges, for structural strength, and the trailing edges, for attachment points for control surfaces. These stringers were sanded to the shape of the desired airfoils. Basswood ribs are secured by wooden dowels, which also increases rigidity. The vertical and horizontal stabilizers were constructed separately, then joined to form the tail. Following electronic integration, the tail was covered in UltraCote. The tail construction is shown in Fig 6.6 [1].

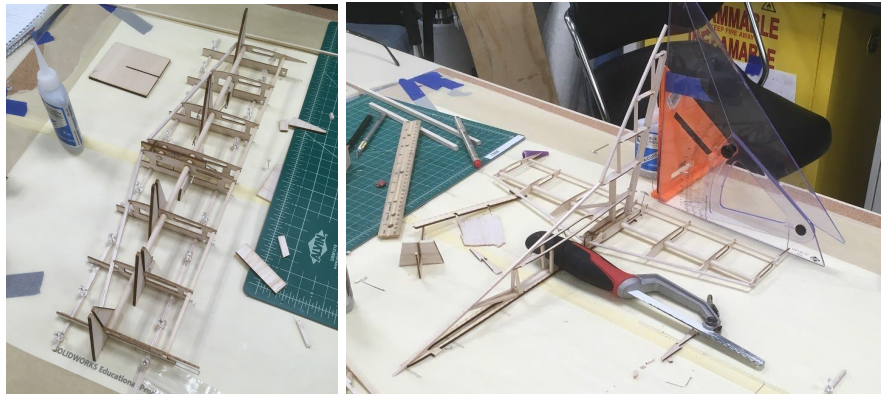


Figure 6.6: Left: horizontal stabilizer construction. Right: horizontal and vertical stabilizers [1].

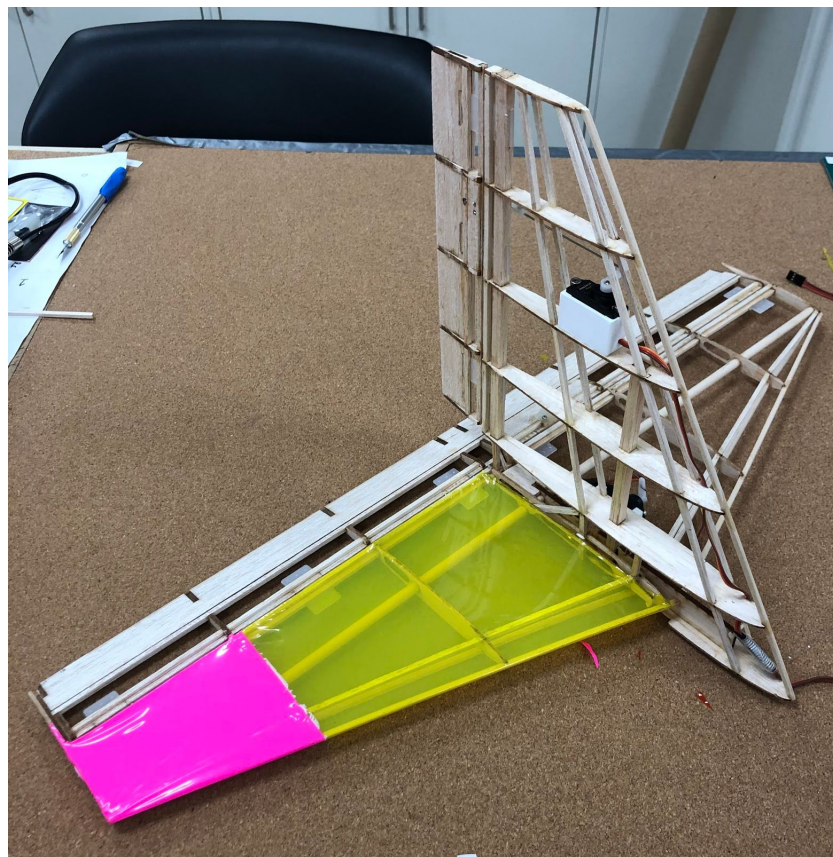


Figure 6.7: Current status of 2020-2021 tail.

6.2.7 Sensor Mechanism

The sensor release mechanism is to be constructed using a servo mechanism to launch the sensor, attached via a string, from its housing bay so that the string becomes taut when the Bearon is in flight. The sensor will preliminarily be recovered by reversing the servo connected to a spool, assisted by a ramp at the entrance to the cargo bay.

6.2.8 Sensor

The sensor consists of a 3D-printed cylindrical body with internal weights. For stabilization, four fins are added to the end of the sensor. Several detachable nose profiles will be tested, with conical, elliptical, and plate profiles. The tow line will be attached directly to the cylindrical body, through the nose cone for safety. Sensors will be stored in the compartment that was used for passengers in the *Atlas*, which can fit 3 sensors total, a fourth will be stowed below the compartment.

6.3 Manufacturing Milestones

The manufacturing schedule was implemented to ensure components were completed and tested on time. The prototype and the final aircraft were divided into important subtasks, as shown in Fig. 6.8.

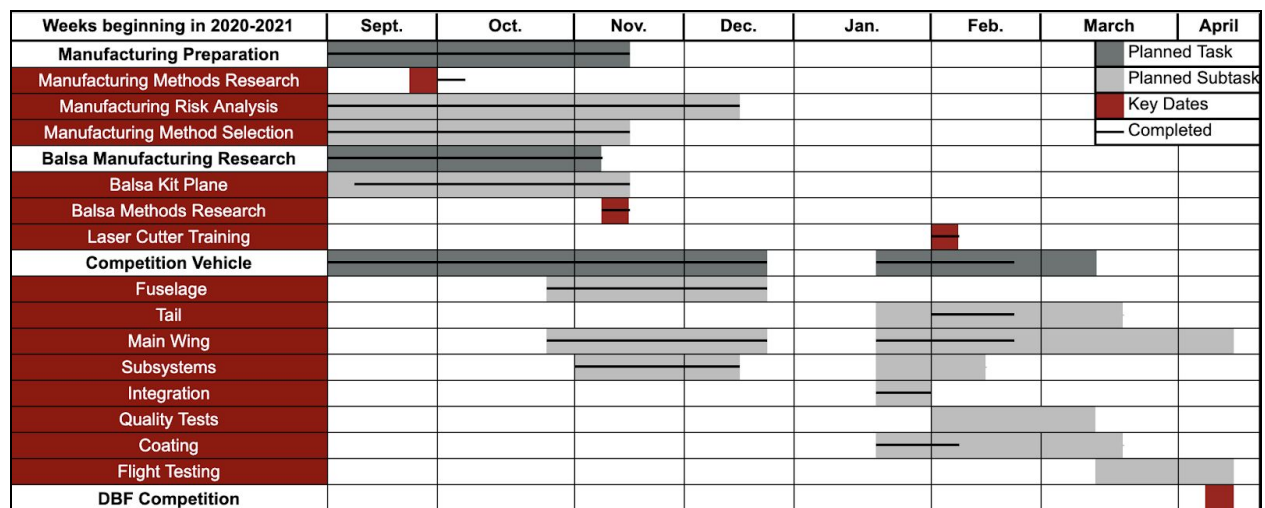


Figure 6.8: Gantt chart of the manufacturing plan, showing actual, projected and planned schedule.

7. Testing Plan

7.1 Propulsion Testing

Static propulsion testing was performed on the chosen propulsion package with an RCbenchmark Series 1585 dynamometer mounted on a thrust stand. Using RCbenchmark software, thrust, RPM, current draw, and electrical power measurements were collected at incremental throttle levels. The data was then used to validate the predicted performance described in Section 4.4 and Section 4.5 and to confirm the choice of propulsion package.

7.2 System and Subsystem Testing

7.2.1 Sensor deployment and recovery mechanism

In order to test the release and recovery of the sensor, we will build a bench model to be tested out the side of a convertible for reliable function. We will collect the data from release and iterate the

design to ensure reliability and stability. This mechanism will be tested by retracting the sensor at various angles and with added weight to simulate drag.

7.3 Manufacturing Testing

Since the overall design of the aircraft was not significantly changed this year, and due to covid related limitations, the team decided to rely on previous testing results instead of repeating the tests with the current plane.

Manufacturing testing was performed to ensure that various components could be built and that manufacturing methods were feasible before they were implemented on the aircraft. Manufacturing testing was also performed to verify the integrity of components once they had been incorporated into the aircraft. This two pronged approach was intended to filter serious manufacturing issues early and to confirm that component manufacturing met quality standards as a whole. These tests were designed over the course of the manufacturing process as new components and issues arose [1].

7.3.1 Balsa Wrap Test

The team needed to test whether the main wing leading edge could be covered with balsa, and to test several wrapping methods. To accomplish this, small sections of the wing were laser cut and assembled. Balsa sheets were soaked in water until adequately flexible and were then wrapped around the leading edge of the wing segment. The sheet was bent to the shape of the airfoil, first without applying leading edge pressure, then while applying leading edge pressure. The third method tested was pre-bending the sheet on a wider circular rod before applying pressure to the leading edge and then bending the sheet to the shape of the airfoil, to make bending easier. The team tested methods to hold sheets in place while drying, including holding the sheet in place by hand, using rubber bands to secure the sheet, pinning the sheet (pictured in Fig. 8.1), and using the heat gun to decrease drying time [1].

7.3.2 Trailing Edge UltraCote Test

The trailing edges of the main wing, Fowler flaps, and ailerons were not strong enough to withstand the pressure exerted by the UltraCote when heat-shrunk. To find the simplest effective method, small sections of the wing were laser cut, assembled using various reinforcement methods described in Table 7.1, and coated.

Table 7.1: Reinforcement methods for the trailing edge [1].

Reinforcement Method	Method
None	Ribs with no connector between trailing edges
Toothpick	Toothpick cross-bracing
Thin Balsa	½ in. balsa sheets sanded to match trailing edge
Medium Balsa	1 in. balsa sheets sanded to match trailing edge
Multigrain Balsa	Medium Balsa Reinforcement and rib bracing in 2 directions

7.3.3 Flutter Test

The team performed a torsion test on the tail to gauge flutter risk. The team could not be sure how rigidly attached the tail would be until both it and the fuselage were built and attached. At that stage, the team was unable to perform any analysis on the tail. As a result, the team measured whether the stiffness of the tail was broadly acceptable or broadly unacceptable based on past experience [1].

7.3.4 Wingtip Test

The team performed a wing tip test. To perform the test the team lifted the plane with multiple people supporting the wing and then slowly removed support until just the wingtips were supported. One member also placed a hand on the tail to prevent the plane from pitching downwards [1].

7.3.5 Bad Weather Test

The team planned to perform a bad weather test on last year's prototype aircraft to ensure that the aircraft could cope with potential wind and rain in Wichita. The team brought the aircraft outside in both light and moderate rain. This tested the water resistance of the aircraft, how water interacted with holes in the UltraCote for the landing gear and servo control arms, how water affected Kapton tape, and how the aircraft performed in inclement weather [1].

7.4 Data Collector Testing

The Pixhawk, GPS sensor, and pitot tube were first tested indoors to determine data collection procedures. Next, the GPS satellite lock was tested outdoors and accuracy confirmed by walking around with the GPS. The calibration process was tested several times to ensure all values were initialized correctly. Before each test flight, the system will be calibrated with QGroundControl and tested to ensure the estimated states are accurate. After each flight, the data, recorded with an SD card, is uploaded to Flight Review. Filtering with an extended kalman filter is built in. Produced graphs and maps are sanity-checked to ensure the data is recording correctly and is free of errors [1].

7.5 Wind Tunnel Testing

The wind tunnel plan consists of three phases: model construction, data collection, and data analysis. For initial testing purposes, a single 3-D printed model served as the *Red Bearon* model. This half-span, 4:15 scale model was mounted in the wind tunnel test region with a variable angle positioning system and three-component pistol grip sting balance. The mount was fitted to position the fuselage inside the boundary layer of the test region. The team calculated lift and drag characteristics for the proposed aircraft from the axial and normal force readings of the pistol grip balance. After accounting for normal and axial forces occurring due to the model weight, sting balance, and mounting device, the team collected data for several AoA. From this data, C_L and C_D were calculated using MATLAB and overlaid against an inviscid analysis in XFLR5 [1].

7.6 Pre-Flight Checklist

Table 7.2: Flight checklist [1].

Component/System	Action to take
Environment	☐ Record atmospheric conditions, wind speed, and wind direction
Batteries	☐ Check propulsion battery is securely connected, mounted, and fully charged ☐ Check receiver battery is securely connected, mounted, and fully charged
Airframe	☐ Perform general shakedown checking servo linkages, hinges, and load bearing paths
Mission Specific	☐ Confirm all sensors are locked in place (M2) ☐ Confirm sensor release mechanism is set up properly (M3)
CG Location	☐ Confirm CG location is within acceptable limits marked on the airframe
Landing gear	☐ Confirm landing gear apparatus is fixed to the airframe and the wear on wheels ☐ Confirm that the aircraft tracks straight
Electrical Subsystems	☐ Perform initialization and confirm data collection for Pixhawk flight data recorder ☐ Check control surface deflections for rate, max deflection, and mapping to control sticks
Propulsion	☐ Check motor, prop direction, and ensure screws/nuts are tight and in good condition ☐ Check throttle power, aircraft vibration, and thrust vectoring system
On Site Safety	☐ Ensure clear takeoff and landing paths ☐ Detail expected flight plan to personnel; ensure all personnel are stationed correctly

7.7 Flight Testing Plan

Flight tests were performed with the first iteration design. Minor adjustments were made to the design after each test flight. The final competition vehicle will take into account all results from the test flights. Table 7.3 details the planned flights of the *Red Bearon*. These flights may be performed by a prototype or the competition vehicle. As of writing, flight tests 1 and 2 were completed for the 2019-2020 competition [1].

Table 7.3: Planned flight tests [1].

Objective	Notes	Flight Test #
Controlled Flight and Landing	Cooper-Harper Score	1
Landing Gear Change Validation	Ensure structural changes improved gear performance	2
In-Flight Fowler Flap Deployment		2
Stall and Recovery Test		2
Bad Weather Flight	It started to hail (happened to be flight test 2)	2
M1 Lap Time Performance		2
Fowler Flap Takeoff	Confirm aircraft stability with Fowler flaps during takeoff	3
M3 Banner Performance	Deploy banner in flight	3
Takeoff Performance Validation	Full throttle with ailerons and Fowler flaps deployed	4
M2 Max Payload Performance	Carry all 4 sensors	5

8. Performance Results

8.1 Propulsion Testing Results

Static thrust testing was performed on the propulsion package described in 5.3.10. The selected Sunnysky X4125 V3 420 K_v motor fitted with a APC 15x10E prop was attached to the thrust stand and powered by a 6S LiPo Tattu 4500 mAh battery. Using the RCbenchmark Series 1585 dynamometer, thrust, RPM, current draw, and electrical power measurements were collected. The comparison between the predicted performance described in Section 4.4 and the performance determined via this propulsion testing is shown in the table below.

Table 8.1 Comparison of predicted performance to propulsion test results (per motor).

Parameter	Prediction	Test Result
RPM	8000	8396
Thrust [Kgf]	5	5.29
Max Current [A]	75	77.89
Electrical Power [W]	1150	1271

Table 8.1 shows that the performance generally matches the performance predicted in Section 4.4. This validates the choice of propulsion package. Note that since the *Red Bearon* was designed with a two motor configuration, the total thrust, max current, and power are double what is shown in Table 8.1.

8.2 System and Subsystem Testing Results

8.2.1 Sensor deployment and recovery mechanism

Sensor mechanism testing has not been completed at this time.

8.3 Manufacturing Testing Results

8.3.1 Balsa Wrap Test

A summary of the results of the testing of balsa wrapping is shown in Table 8.2.

Table 8.2: Results of Balsa Wrap Tests [1].

Leading Edge Wrapping Method	Result
No Leading Edge Pressure	Sheet split along leading edge
Leading Edge Pressure	Success, with some initial difficulty achieving the desired curvature
Pre-Bending and Leading Edge Pressure	Success, with slight undesired warping from pre-bending
Pre-Bending with 3D printed mold	Success, requires less people than above method
Leading Edge Drying Method	Result
Hand Held	Not viable due to long drying times
Rubber Bands	Success, unable to scale to full wing
Pins (Fig. 8.1)	Generally successful with some tearing, unable to scale to full wing
Heat Gun	Success, must use low setting to prevent injury

This test indicated leading edge pressure as the preferable method. At scale, rubber bands could break the trailing edge and cannot run lengthwise. Basswood stringers proved too strong for the pins to

be used on the wing. Thus the heat gun and hand held methods were used to dry the wood before glue was applied. This test successfully showed that wrapping the leading edge in balsa was feasible [1].



Figure 8.1: Pinned leading edge [1].

8.3.2 Trailing Edge UltraCote Test

A summary of the results of testing UltraCote on the trailing edges is shown in Table 8.3.

Table 8.3: Results of Trailing Edge Test [1].

Reinforcement Method	Result
None	Failure of ribs at trailing edges
Toothpick	Deformed trailing edge due to lack of flat reinforcement
Thin Balsa	Severe warping of trailing edge reinforcement
Medium Balsa	Moderate warping of trailing edge reinforcement
Multigrain Balsa	Minimal/no warping of trailing edge

The multigrain balsa reinforcement method best eliminated warping in the trailing edge. Therefore, variations of this method were employed in the main wing, Fowler flaps, and ailerons [1].

8.3.3 Flutter Test Results

The team determined that the initial rigidity of the tail was insufficient and that the flutter risk was high. The team predicted that adding balsa cross bracing in the tail would increase rigidity to satisfactory levels. After making those changes, the aircraft met rigidity expectations, and flutter risk was reduced [1].

8.3.4 Wingtip Test Results

The team successfully completed a partially loaded aircraft wingtip test. The test was performed with battery weight; however, one motor was missing and no passengers were loaded. The test showed additional wing reinforcements were needed. In the next iteration, the tip chord will increase, allowing the CF rod to span the entire wing and plywood supports will be added [1].

8.3.5 Bad Weather Test

This test confirmed UltraCote is water resistant. The coating performed well in rain and a wintry mix. Some water entered servo holes — the holes will be shrunk. The Kapton tape remained dry. The plane successfully flew in a wintry mix. As such, the design is adequate for bad weather operations [1].

8.4 Wind Tunnel Testing Results

Data from the wind tunnel testing was used to create more realistic simulations. Lift and drag data generated in XFLR5 accounts only for induced drag and ignores fuselage effects and therefore

underestimates total lift and drag. The experiment accounted for both parasitic and induced drag. Figure 8.2 shows experimentally determined and theoretical C_L and C_D for an AoA sweep. The wind tunnel testing setup is shown in Fig. 8.3. Compared to the theoretical results, lift is higher due to fuselage lift [1].

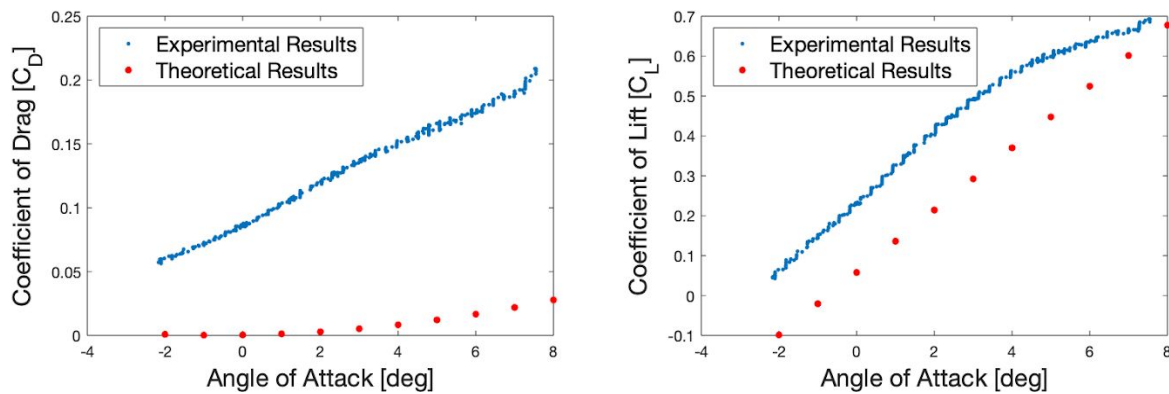


Figure 8.2: Left: C_D for several AoA. Right: C_L as a function of AoA [1].

The tunnel control speed was 107 mph. Matching Re yielded 12 m/s for the full-scale aircraft. Wing area was used as reference for both C_L and C_D . The C_L is reasonable when compared to the analytical results due to the large fuselage's added lift. C_D is higher than expected even when estimating parasitic drag. This difference is the result of mounting device surface area and surface roughness of the 3D printed model. Results produced were consistent and replicable. Lift values validated XFLR5 results. Drag values informed KM Sim drag models [1].

To improve KM Sim, wind tunnel lift data replaced XFLR5 data, accounting for the additional lift from the fuselage. A straight section of flight, flown at constant throttle and airspeed, was modeled in KM Sim and drag values were tweaked so that KM Sim results matched flight test results for airspeed, alpha, and throttle setting. With wind tunnel and motor test data, KM Sim accurately matched flight data allowing WUDBF to produce highly accurate performance estimates [1].



Figure 8.3: Wind tunnel apparatus and setup [1].

8.5 Flight Testing Results

WUDBF completed two successful flight tests of the *Atlas* in 2020. After a receiver failsafe and

range check, the preflight checklist (Table 7.2) was completed. During taxi testing, the aircraft experienced catastrophic gear failure. WUDBF switched the landing gear structure to birch plywood and increased the number of cross sections connecting the landing gear platform to the carbon rod [1].

On 1/17/2020 WUDBF completed the first successful flight of the *Atlas*. The plane tracked well on takeoff but, as velocity increased, the *Atlas* encountered a vibrational mode due to an impulse force. The pilot avoided further complications by quickly increasing throttle, and plans to increase damping within the landing gear structure will solve this problem. The *Atlas* has good handling characteristics considering its high wing loading. Once trimmed, the plane tracked straight with ample elevator and roll authority. Coordinated turns were required and the pilot felt the vertical tail was undersized. WUDBF will increase its size on the competition vehicle. The pilot noted the rudder control was sensitive. Overall, the aircraft earned a Cooper-Harper score of 3. The aircraft cruised at 20.8 m/s and traveled 6.42 km in 4:15 minutes, reaching a maximum altitude of 137.5 m and a maximum loading of 3.8g. Figure 8.4 details the aircraft's airspeed over time. In response to a landing gear failure on landing, WUDBF increased gear rigidity by shifting the attachment mechanism from a square carbon rod to a plywood sheet between main fuselage bulkheads [1].

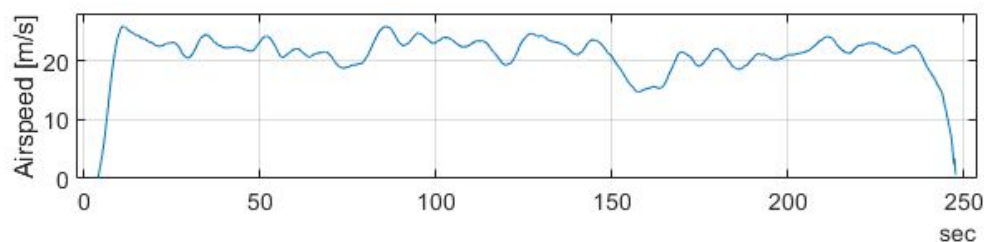


Figure 8.4: Airspeed of the *Atlas* in its maiden flight test [1].

WUDBF tested the *Atlas* in a second successful flight on 1/31/2020. In this 5:28 flight, the *Atlas* flew 8.23 km, averaging 19.2 m/s with a maximum altitude of 118 m. Figure 8.5 shows the aircraft trajectory. In-flight Fowler flap testing was performed. The pilot found Fowler flap deployment to induce minimal negative pitching moment, easily compensated for using elevator input. The aircraft experienced controllable benign stall characteristics dropping a wingtip slightly at the very end of the stall. However, due to the downwash built in at the wingtips, the pilot maintained post stall roll control. The *Atlas* has high wing loading, resulting in stall speeds of 13.6 m/s and 9.1 m/s without and with flaps, respectively (See Table 5.7). This requires difficult, high velocity landings. The plane landed with an unequal mass distribution on the main gear, causing the aircraft to roll. This resulted in minor damage to the tail and motor mounts. WUDBF plans to widen the gear base on the competition vehicle for stability and further strengthen the mounting system [1].

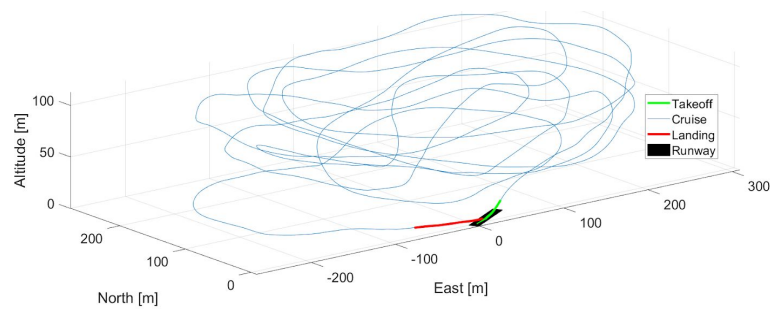


Figure 8.5: Trajectory plot of the *Atlas* using GPS data [1].

Finally, after the construction of the competition vehicle is completed, WUDBF will perform similar system flight checks, ensuring a safe and operable competition vehicle.



Figure 8.6: Takeoff climb of the *Atlas* on a cold winter morning [1].

9. Bibliography

- [1] Washington University in St. Louis Design/Build/Fly Team, "The Atlas," *AIAA Design/Build/Fly 2019-2020 Design Report*, retrieved 10 February 2021.
- [2] Prandtl, L., "Applications of Modern Hydrodynamics to Aeronautics," NACA-TR-116, Jan. 1923, pp. 190.
- [3] Bowers, A. H., Murillo, O. J., Jensen, R. R., Eslinger, B., and Gelzer, C., "On Wings of the Minimum Induced Drag: Spanload Implications for Aircraft and Birds," NASA/TP-2016-219072, Mar. 2016.
- [4] Abbott, I. H. and von Doenhoff, A. E., *Theory of Wing Sections*. Dover Publications, Inc., New York, 1959.
- [5] Traub, L. W., and Kaula, M. P., "Effect of Leading-Edge Slats at Low Reynolds Numbers," *Aerospace*, Vol. 3, No. 39, 2016, pp. 3-7. doi:10.3390/aerospace3040039
- [6] Caughey, D. A., "Introduction To Aircraft Stability and Control Course Notes for M&AE 5070," Lecture Notes, Sibley School of Mechanical & Aerospace Engineering, Cornell University, Ithaca, New York, 2011, pp. 25-101.
- [7] Foster, T. M., and Bowman, J., "Dynamic Stability and Handling Qualities of Small Unmanned-Aerial-Vehicles," AIAA Paper 2005-1023, Jan 2005. <https://doi.org/10.2514/6.2005-1023>.
- [8] Durmus, S., "Effects of Cubic Wing Loading Parameter on Airplane Wing Sizing and Parasitic Drag," *J Aeronaut Aerospace Eng*, Vol. 9, No. 226, 2020 pp. 2-5.
- [9] "The *Watts Per Pound* rule," RC Airplane World, retrieved 15 February 2021. <https://www.rc-airplane-world.com/watts-per-pound.html>